



Equity–bond correlation regimes and dynamic allocation conditioned on macroeconomic factors

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¹All source code for this project is available at <https://github.com/EnsaStatApp/CorrelationEquityBond>.

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1 Introduction

The correlation between equities and bonds is one of the most structurally important parameters in asset allocation: it determines a portfolio's diversification benefits and, consequently, its overall risk profile. Yet far from being stationary, this correlation has undergone major reversals over the past fifty years — positive during the stagflation of the 1970s, durably negative after the year 2000, and positive once again during the post-COVID inflationary cycle of 2022–2024. These shifts reflect deep changes in the macroeconomic environment and call into question any static modelling of the dependencies between asset classes. This correlation, however, cannot be studied in isolation but rather jointly with that of volatilities: the negative correlation observed after 2000 is not the same as the one prevailing during the 2008 financial crisis. It is therefore natural to assume the existence of distinct market phases — referred to as *regimes* — in which assets display different risk dynamics. The challenge then lies in modelling these regimes and quantifying their transitions.

This project addresses this instability as an applied statistics problem. The aim is to build and validate, on the monthly returns of the S&P 500 and the 10-year Treasury Bond from 1970 to 2025 (Shiller, 2018), a probabilistic latent-regime framework based on Hidden Markov Models (HMM) capable of representing this non-stationarity. An extension with time-varying transition probabilities (*Hidden Markov Model Time-Varying Transition Probabilities*, HMM-TVTP) is then estimated in order to assess the extent to which these transitions are explained by observable macroeconomic covariates.

The question we pose is therefore the following : *can we identify statistically robust latent market regimes whose transitions are driven by macroeconomic factors, and derive from them an allocation that outperforms static or rolling benchmarks over a long cycle ?*

This approach builds on the literature on the probabilistic modelling of financial regimes (Hamilton, 1989; Filardo, 1994; Diebold et al., 1994) and proceeds in two stages. First, we conduct a descriptive analysis at three levels: we first identify latent market regimes from financial returns alone via an HMM; we then estimate pure macroeconomic regimes from FRED indicators (Federal Reserve Bank of St. Louis, 2026) in order to assess the extent to which a macroeconomic regime implies a distinct financial regime; and we finally combine these two perspectives to obtain market regimes whose transitions are driven by the macroeconomy. Given the instability inherent in these models, their statistical validity is assessed through a composite score aggregating separability, numerical stability, persistence, and detection conviction. Second, we propose a predictive approach in the form of a dynamic allocation evaluated over 1995–2025 on an *expanding* window and compared against standard benchmarks. The results are finally illustrated through two case studies: the 2008 financial crisis and the 2022 inflationary crisis.

The report is organised as follows. Section 2 presents the literature review. Section 3 documents the non-stationarity of the series. Section 4 sets out the probabilistic modelling framework. Section 5 presents the empirical results of the three successive descriptive approaches (market regimes, pure macroeconomic regimes, and market regimes conditioned on the macroeconomy). Section 6 describes the dynamic allocation strategy and its performance. Section 7 concludes.

2 Literature review

2.1 Instability of the equity-bond correlation and macroeconomic drivers

The non-stationarity of the equity–bond correlation is well documented in the literature. Historically, this correlation remained predominantly positive between the 1970s and the 1990s, in an environment marked by supply shocks and high inflation. It subsequently turned negative from the bursting of the dot-com bubble onwards, with bonds increasingly playing the role of a safe-haven asset. This negative regime, which long supported diversification strategies, was nonetheless called into question during the inflationary cycle of 2022–2024, a period in which equities and bonds declined simultaneously.

Czaronis et al. (2021) highlight the unstable nature of this correlation and show that extrapolating it from historical monthly data is unreliable for long-term horizons. Indeed, the presence of autocorrelations and lagged cross-correlations implies that the correlation depends heavily on the horizon considered. To address this limitation, the authors introduce the notion of *single-period correlation*, which aims to better capture the cumulative co-movement of returns. They further identify four main determinants: economic growth, inflation, the relative return between equities and bonds, and their relative volatility. Their approach, based on filtering historical observations according to the current context, substantially improves predictive power relative to traditional methods.

This work suggests a close link between the macroeconomic regime and the correlation between assets, which this dissertation seeks precisely to formalise. In the same vein, Brach (2025) shows that classical dynamic estimation methods (rolling windows, EWMA) remain essentially reactive: they capture breaks at the cost of strong instability, whereas long-memory methods smooth structural changes excessively. This tension justifies the recourse to regime-based models, capable of detecting sharp rather than gradual breaks.

Beyond correlations, the autocorrelation of returns itself can bias standard risk measures: Burghardt and Liu (2012) show that the naive square-root-of-time rule underestimates risk in the presence of positive autocorrelation, while Petreski (2007) documents the amplification of this bias in *mark-to-model* valuation practices on illiquid assets.

2.2 Probabilistic modelling of financial regimes

Hidden Markov Models (HMM), introduced by Hamilton (1989) in the analysis of economic cycles, rest on the idea that financial series are generated by several underlying processes whose selection is governed by an unobserved Markov chain. Each regime is thus associated with a specific dynamic of return, volatility, and correlation. In their standard version, transitions between regimes are characterised by fixed probabilities.

The main appeal of this approach lies in its ability to identify abrupt breaks without imposing, *a priori*, either their number or their dates. It also makes it possible to associate each observation with a probability of belonging to a regime, which offers an interpretable economic reading. In this respect, it differs from approaches based on rolling windows or GARCH-type models.

The extension to time-varying transition probabilities (*Time-Varying Transition Probabilities*, TVTP), developed notably by Filardo (1994) and Diebold et al. (1994), introduces a dependence of these probabilities on observable macroeconomic variables, such as inflation, interest rates, or credit spreads. This extension is particularly relevant to the present work, since it makes it possible to test explicitly whether

transitions between market regimes are influenced by the macroeconomic environment. It thus paves the way for a dynamic allocation conditioned on the available macroeconomic information.

2.3 Inflationary regimes and asset behaviour

The relationship between inflation and asset returns also appears strongly regime-dependent. Bouzida and Brach (2019) identify several inflationary regimes in the United States over the 1970–2018 period, using statistical classification methods based on the returns of cash, oil, and gold. Their results show that long-term average correlations between assets and inflation mask substantial heterogeneity across the regimes considered.

In particular, assets that provide a good hedge in a supply-shock context are not necessarily the most effective under conditions of excess demand. The authors also stress that certain episodes linked to unconventional monetary policies fall outside classical regression-based analytical frameworks. This reinforces the value of models capable of endogenously detecting breaks in the relationships between variables.

Ultimately, the literature converges on the idea that correlations between asset classes are unstable, dependent on the economic regime, and difficult to apprehend through classical smoothing methods. In this context, HMM models and their TVTP extension appear as particularly well-suited tools for identifying robust latent regimes and analysing the role of the macroeconomy in their transitions, which constitutes the core of this work.

3 Empirical evidence of non-stationarity and financial regimes

3.1 Descriptive statistics and structure of the financial and macroeconomic data

The analysis is based on a monthly sample of 667 observations beginning in 1970, a pivotal period that captures several contrasting cycles of inflation and growth. The financial assets selected, the S&P 500 (SP) and U.S. 10-Year Treasuries (T10), are transformed into *Log Total Returns* in order to stabilise the distributions.

Table 1: Descriptive statistics of the market variables and statistical fit (1970-2025)

Variable	Mean	Median	Std. dev.	Nu (ν)	LL Normal	LL Student
SP Log Return	0.86%	1.19%	3.69%	4.02	1255.05	1304.75
T10 Log Return	0.53%	0.55%	2.02%	5.60	1656.00	1681.40

Table 1 presents the descriptive statistics and goodness-of-fit tests for these returns. We observe the classic return-risk hierarchy: equities display a higher average monthly return (0.86%) but at the cost of a volatility nearly twice as high as that of bonds (3.69% versus 2.02%).

In order to make the asset series compatible with the assumptions of our models, we compare the fit of a normal distribution and a Student distribution using the log-likelihoods (LL). This procedure aims to identify the distribution that best describes the fat tails observed in financial data. The results indicate that the Student distribution provides a better fit, with systematically higher LLs (notably 1304.75 versus 1255.05 for the S&P 500). The estimated degrees of freedom (ν) are particularly low for equities ($\nu \approx 4.02$), which reflects very pronounced distribution tails and a risk of extreme downside moves.

However, within the framework of our hidden Markov chain models and asset allocation, the normality assumption remains simpler to handle and allows closed-form analytical solutions to be obtained. For this reason, we adopt a Gaussian specification for the remainder of this study, while remaining aware that this simplification could be relaxed in subsequent extensions of the model.

The macroeconomic variables are transformed to be made stationary (no missing data being present in the retained sample), in line with the standard assumptions of time-series models. To verify the validity of these transformations, we use the Augmented Dickey-Fuller (ADF) stationarity test, which detects the presence of a unit root in a time series.

To examine the relationships between the selected macroeconomic variables, we present the correlation matrix in Figure B.1 of the appendix. Several notable relationships emerge, in particular between the *Credit Spread* and the slope of the yield curve (*Slope*), as well as between the industrial activity indicators (*Log INDPRO diff* and *Log PAYEMS diff*). The study of these interdependencies highlights significant collinearities, making it possible to identify variables that move in a similar way and to anticipate potential problems of redundancy or multicollinearity in the empirical models.

To synthesise the information contained in these variables and identify the main macroeconomic factors, we apply a Principal Component Analysis (PCA), the results of which are presented in Figure B.2. This factor-based approach is justified by the dependence structure observed above. The variables are first made stationary (see other project notes). The first three components explain approximately 67.5% of the total variance, which is consistent with the observed correlation structure. The examination of the factor loadings suggests an economic interpretation of the main components: PC1 is mainly associated with the industrial activity indicators, PC2 with inflation dynamics, and PC3 with variables related to monetary conditions.

3.2 Empirical evidence of non-stationarity

The examination of the historical dynamics of the assets invalidates any approach based on a constant correlation. The analysis presented in Figure 1 relies on an estimation by exponentially weighted moving averages (*EWMA*) with a *span* of 24 months. This method captures inflection points more responsively than a standard rolling average while stabilising the estimates.

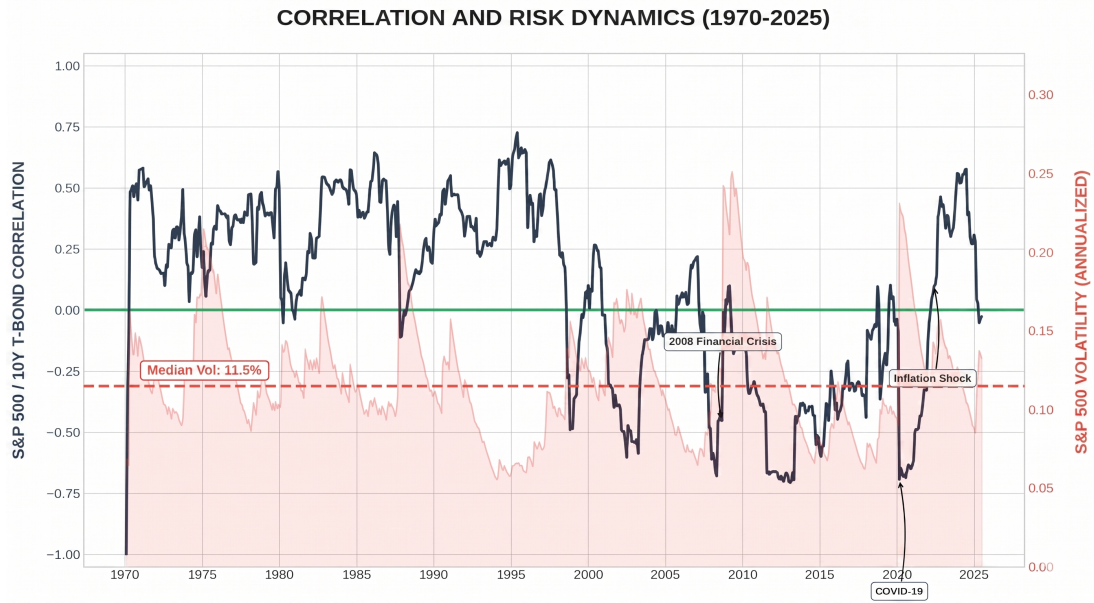


Figure 1: Joint dynamics of the correlation (EWMA) and the volatility of the S&P 500 (1970-2025)

The chart highlights a chronic instability of the return-risk pair. We observe that the correlation between equities and bonds oscillates erratically between phases of strong positive dependence (close to +0.75) and periods of marked inversion (close to -0.75). The neutrality line (in green) is frequently crossed, indicating that the diversification capacity of bonds relative to equities is not an intrinsic property, but rather a characteristic that is unstable over time. Visually, these fluctuations trace out zones of contrasting behaviour, where the persistence of a positive correlation suggests a weakening of the protective role of bond assets, while shifts into negative territory mark episodes of increased diversification.

In parallel, the volatility of equities (shaded area) shows stress peaks that deviate significantly from the historical median of 11.5%. It appears visually that the sharpest correlation breaks often coincide with these phases of heightened risk. This co-dependence between the level of risk and the correlation structure confirms the non-stationarity of the market parameters. It justifies the recourse to a probabilistic modelling approach capable of capturing these state changes endogenously.

4 Probabilistic modelling of financial and macroeconomic regimes

4.1 Theoretical framework: hidden Markov models

Hidden Markov models assume that the observations are generated by a latent stochastic process evolving between K discrete regimes, denoted $z_t \in \{1, \dots, K\}$. We present here the two variants used in this report: the homogeneous HMM and its version with time-varying transition probabilities (HMM-TVTP).²

²The following presentation assumes the reader is familiar with stochastic calculus, financial modelling, and standard statistical tests. The detailed mathematical developments (EM algorithm, Viterbi decoding) are deferred to Appendix A.1, and the entire logic of the statistical tests and their application to HMMs to Appendix A.2.

4.1.1 Homogeneous HMM

The *Hidden Markov Model* (HMM) assumes a first-order Markov chain with a constant transition matrix:

$$[P]_{ij} := p_{ij} = \mathbb{P}(z_{t+1} = j \mid z_t = i), \quad p_{ij} = \frac{\exp(\beta_{ij})}{\sum_k \exp(\beta_{ik})}$$

where β_{ij} is the *intercept* of the transition $i \rightarrow j$ (a softmax parametrisation guaranteeing $p_{ij} \in [0, 1]$ and $\sum_j p_{ij} = 1$). The emissions are multivariate Gaussian conditional on the regime:

$$O_t \mid z_t = k \sim \mathcal{N}(\mu_k, \Sigma_k),$$

the observations being assumed independent conditional on z_t . The parameters break down into emissions $\theta_E = (\mu_k, \Sigma_k)_k$ and transitions $\theta_T = (\beta_{ij})_{i,j}$. Maximum-likelihood estimation relies on the Baum-Welch algorithm (an EM variant for latent variables), the most likely sequence of regimes then being obtained via Viterbi. The algorithmic details are given in Appendix A.1.

4.1.2 HMM-TVTP

The *HMM with Time-Varying Transition Probabilities* (HMM-TVTP) relaxes the homogeneity assumption: the transition matrix depends on a vector of covariates X_t (macroeconomic variables in our case):

$$p_{ij}(X_t) = \frac{\exp(\beta_{ij} + \omega_j^\top X_t)}{\sum_{k=1}^K \exp(\beta_{ik} + \omega_k^\top X_t)}$$

where ω_j measures the impact of the covariates on the transition towards state j (independently of the originating state, to avoid overfitting). The transition parameters are augmented to $\theta_T = (\beta_{ij}, \omega_j)_{i,j}$, and the estimation relies on a generalised EM procedure in which the M-step is solved numerically by quasi-Newton (L-BFGS). The algorithmic details are given in Appendix A.1.

More generally, HMMs make it possible to model temporal dynamics with hidden states. In this report, we use the `ssm` library for the implementation and estimation of the models, which allows discrete latent dynamics to be handled efficiently without entering into the general formalism of State-Space Models ([SSM GitHub Repository](#)).

4.2 Regularisation and hyperparameters

The number of regimes K is selected through information criteria (AIC, BIC) and economic validation of the resulting states. A Dirichlet prior is applied to each row of the transition matrix:

$$\vec{\alpha}_k = \alpha \vec{1} + \kappa \vec{e}_k,$$

where α regularises the set of transitions and κ specifically reinforces persistence ($k \rightarrow k$ transitions). For the HMM-TVTP, an L_2 penalty is also applied to the coefficients ω_j .

To ensure the reliability of the identified regimes and avoid overfitting biases, each model is evaluated according to a global statistical robustness score s_G aggregating statistical separability, numerical stability, state persistence, and detection conviction. The mathematical detail of these validation indicators is

reported in Appendix A.2.

5 Empirical results

5.1 Latent market regimes

A first approach to modelling risk regimes can be to observe the log returns of financial assets and thereby obtain market regimes. For this, we used the S&P and the *10-year Treasury Bond* from 1970 to 2025. This analysis therefore remains, for now, purely descriptive and not predictive.

5.1.1 Statistical justification of the hyperparameters

First of all, the number of regimes K is a central hyperparameter of hidden Markov models. A K that is too low risks failing to capture the diversity of market dynamics, whereas a K that is too high may lead to overfitting and to regimes that are difficult to interpret economically.

To select an appropriate K , the model was estimated for several values of K , and the standard information criteria, in particular the *Akaike Information Criterion* (AIC) and the *Bayesian Information Criterion* (BIC), were compared. These criteria penalise model complexity while rewarding a better likelihood.

The analysis, illustrated by Figure C.1 in the appendix, shows the classic trade-off between parsimony and goodness of fit. Although the BIC suggests $K = 2$, this criterion tends to underestimate the complexity of low-frequency financial series. The AIC, by contrast, decreases up to $K = 5$, indicating that an overly simple model would fail to capture important dependence structures. The choice of $K = 4$ thus corresponds to an optimal compromise: economically, it makes it possible to represent the four states resulting from the combinations of volatility (high/low) and correlation (positive/negative). The other hyperparameters, the regularisation coefficients κ and α , are set to standard values.

5.1.2 Characterisation of the regimes (emission parameters)

The estimation of a homogeneous HMM model with $K = 4$ regimes makes it possible to identify four distinct market dynamics, characterised above all by levels of risk via the covariance matrices between the S&P and the *10-year Treasury Bond*.

Table 2: Characteristics of the HMM market regimes (annualised volatilities in % and 95% Bootstrap CI)

Regime	Freq.	Duration	$\sigma_{S\&P}$	CI _{95%}	σ_{TBond}	CI _{95%}	Correlation	CI _{95%}
0	0.164	5.95	20.77	[16.80 ; 24.42]	6.34	[5.32 ; 7.28]	-0.456	[-0.657 ; -0.221]
1	0.461	21.25	9.48	[8.68 ; 10.27]	6.08	[5.60 ; 6.52]	0.387	[0.304 ; 0.464]
2	0.111	7.36	16.50	[13.34 ; 19.20]	13.57	[10.77 ; 15.93]	0.283	[0.079 ; 0.474]
3	0.263	11.31	7.02	[6.25 ; 7.72]	4.79	[4.24 ; 5.32]	-0.147	[-0.306 ; 0.018]

We can thus draw a first economic analysis:

- **Regime 0 — Flight-to-quality:** a rare and intense crisis regime marked by a sharp fall in equities, high volatility, and a strongly negative correlation, reflecting a shift of investors towards bonds.
- **Regime 1 — Normal regime:** a frequent and classic market phase with moderate volatility and a positive correlation between equities and bonds, suggesting a common influence of macroeconomic factors.

- **Regime 2 — Inflationary stress:** a rare period of high volatility for both asset classes and a positive correlation, typical of macroeconomic environments dominated by inflationary or monetary-policy shocks.
- **Regime 3 — Ideal diversification:** a relatively stable and long market environment characterised by moderate volatility and a slightly negative correlation between equities and bonds, offering diversification benefits.

Furthermore, we can note that the average durations and frequencies of each regime are not too low, which confirms, *a posteriori*, the choice of $K=4$.

In the appendix (see Table C.1), we have presented the Levene and Fisher tests between regimes. A hierarchy in the separation of the regimes thus emerges:

- **By correlation:** The model clearly distinguishes positive-correlation environments (regimes 1 and 2) from negative-correlation ones (regimes 0 and 3).
- **By volatility:** Within pairs with similar correlations, it is volatility that distinguishes the regimes. For example, the difference between the "Normal" regime (1) and "Inflationary stress" (2) stems from the sharp increase in bond volatility ($p < 0.001$), the correlation remaining statistically identical ($p = 1.000$). Likewise, the shift from "Ideal diversification" (0) to "Flight-to-quality" (3) is characterised by an extreme rise in equity volatility ($p < 0.001$).

5.1.3 Dynamics of the regimes (transition parameters)

We obtain the following transition matrix:

$$P = \begin{pmatrix} 83.20\% & 4.99\% & 3.36\% & 8.46\% \\ 2.10\% & 95.29\% & 1.25\% & 1.35\% \\ 3.14\% & 7.57\% & 86.42\% & 2.88\% \\ 5.44\% & 1.95\% & 1.46\% & 91.16\% \end{pmatrix}$$

Table 3: Significance of the transition parameters (Reference: Regime 1)

Transition	Coefficient (vs R1)	p-value	Significance
$R0 \rightarrow R0$	2.8145	1.0×10^{-4}	***
$R2 \rightarrow R2$	2.4354	4.0×10^{-4}	***
$R3 \rightarrow R3$	3.8467	3.0×10^{-4}	***
$R1 \rightarrow R0$	-3.8162	$< 10^{-4}$	***
$R1 \rightarrow R2$	-4.3308	$< 10^{-4}$	***
$R1 \rightarrow R3$	-4.2535	$< 10^{-4}$	***
$R0 \rightarrow R2$	-0.3958	7.287×10^{-1}	
$R0 \rightarrow R3$	0.5282	5.983×10^{-1}	

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

First, we find that regime 1 appears as a quasi-absorbing state, with a probability of remaining in the same state of 0.953 and transition probabilities towards the other regimes systematically below 0.02. This seems to suggest that it constitutes the system's reference regime, corresponding to a stable and

structurally dominant market environment.

Table C.4 in the appendix presents the significance of the transition intercepts from a regime i towards a regime j , with regime 1 as the reference state. We observe that for all regimes i , the probability of remaining in it is significantly ($p \leq 0.01$) higher than that of switching to regime 1, which further confirms the stability of the regimes. For example, the coefficient of 2.81 means that P_{00} is roughly $e^{2.81} \approx 16.6$ times larger than P_{01} .

From the table in Appendix C.4, we can also observe that the coefficients associated with transitions from a regime to itself are systematically positive and statistically significant, which confirms, *a posteriori*, the strong persistence of the states. Conversely, inter-regime transitions are predominantly negative and often significant, reflecting a low probability of spontaneous switching between regimes in the absence of an exogenous shock. Moreover, Table 2 and Appendix C.8 establish a clear hierarchy between persistent, stable regimes and others that are more unstable and rare. Indeed, regime 1 appears as the most stable (average duration of 20 months), the most frequent (46%), and with the highest persistence probability (95%), which makes it the system's reference state. Conversely, regime 0 displays a more transitory dynamic, with a lower average duration (6 months), and is rarer (16%).

5.1.4 Temporal evolution of the regimes and the correlation

In Figure 2, the evolution of the cumulative wealth index, segmented by the Hidden Markov Model (HMM), reveals a strong correspondence between the identified states and the macroeconomic chronology of the S&P 500 since 1970. The *Inflationary stress* regime captures the stagflation periods of the 1970s as well as the 2022 monetary pivot. In parallel, the *Flight-to-quality* phases systematically isolate exogenous shocks and systemic crises (1973, 2008, 2020), characterised by a sharp rise in volatility and a negative correlation. This persistence of the states, particularly visible during the "Great Moderation" (1985-2000) in the *Normal* regime, validates the model's ability to discriminate between non-linear market dynamics beyond a simple trend analysis. Furthermore, Figure 3 represents the correlation between the S&P and the 10-year T-Bond. We can note a shift towards negative-correlation regimes after 2000. Indeed, we can recover the 2008 *flight-to-quality*, when bonds served as a safe haven for investors. Conversely, we can observe a shift towards positive-correlation regimes shortly before 2022, when both the S&P and the T-Bond saw their returns decline.

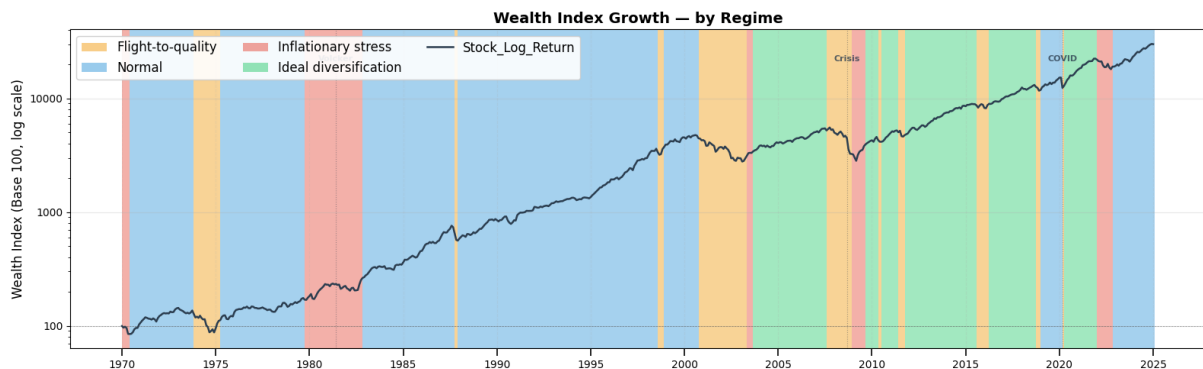


Figure 2: S&P - Cumulative return by HMM market regime

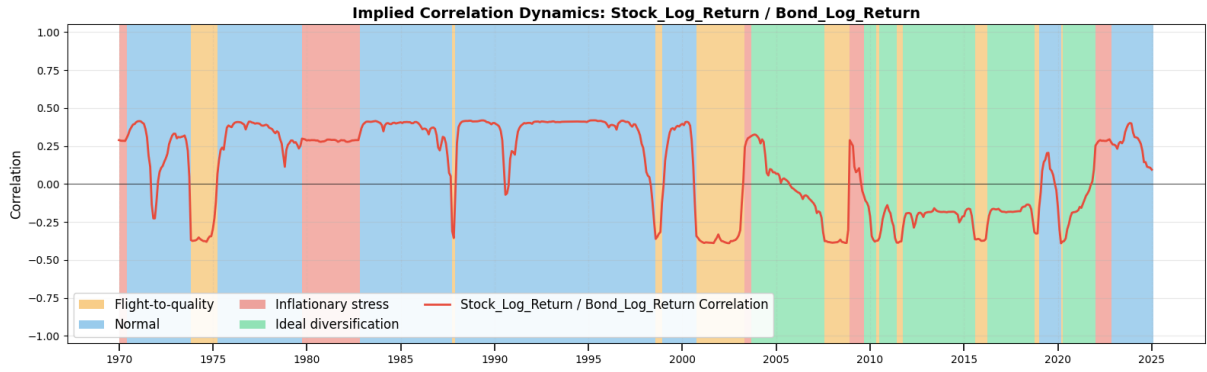


Figure 3: S&P - Correlation between the S&P and the 10-year T-Bond

5.1.5 Limitation of this first model

In sum, this model does succeed in distinguishing the different states of the S&P/T-Bond market. However, it makes it possible to establish neither explanatory links nor predictions with the macroeconomy. Indeed, in this topic, we would like to quantify links between macroeconomic variables such as inflation and the S&P/T-Bond correlation.

Consequently, we decided to carry out a second approach that places the macroeconomy at the heart of the regimes, in order to model links between the macroeconomy and market risks.

5.2 Pure macroeconomic regimes

This second approach tackles the problem from a different but related angle, in order to answer a central question: does a macroeconomic regime always imply a financial regime? To answer it, we no longer fit our model on market data but on a set of macroeconomic variables drawn from FRED (Federal Reserve Bank of St. Louis, 2026), presented below.

Table 4: Description of the macroeconomic indicators (Source: Federal Reserve Bank of St. Louis (2026))

Code	Indicator	Description / Unit
INDPRO	Industrial Production	Real output volume index
PPIACO	PPI (All Commodities)	Producer price index (all commodities)
TCU	Capacity Utilization	Industrial capacity utilisation rate
UNRATE	Unemployment Rate	Unemployment rate (in %)
UMCSENT	Consumer Sentiment	Consumer confidence index (U. Michigan)
TB3MS	3-Month Treasury Bill	3-month Treasury bill rate
CPI	Consumer Price Index	Consumer price index (Inflation)

5.2.1 Model parametrisation and characterisation of the regimes

Since the model structure is the same as before, we do not revisit the hyperparameters, with the exception of the number of regimes: we retain $K = 3$, the $K = 4$ simulations being numerically unstable (non-positive-definite Hessian).

The robustness of the fit was checked via a global coherence score $s_G = 83.1\%$ (threshold: 60%), the detail of which is given in Appendix C.12. The separability reaches 66.7%: the three S&P/T-Bond correlation pairs are all significantly distinct across regimes, and the equity volatility clearly discriminates regimes 0 and 1 as well as 1 and 2. The self-persistent transitions are significant for all regimes (Table C.5 in the appendix), and the average durations — from 37 months for the crisis regime to 93 months for the normal regime — confirm the structural nature of the identified states. The decoding conviction reaches 98.4%. Taken together, these criteria validate the model (verdict: *Robust*).

Table 5: Characteristics of the pure macroeconomic HMM market regimes (annualised volatilities in % and 95% Bootstrap CI)

Regime	Freq.	Duration	$\sigma_{S\&P}$	CI _{95%}	σ_{TBond}	CI _{95%}	Correlation	CI _{95%}
0	0.194	37.41	18.07	[13.40 ; 22.53]	7.60	[6.05 ; 9.13]	-0.346	[-0.551 ; -0.134]
1	0.645	93.36	10.01	[8.83 ; 11.19]	6.07	[5.59 ; 6.51]	0.072	[-0.057 ; 0.203]
2	0.161	46.54	14.05	[12.08 ; 15.88]	9.44	[7.66 ; 11.13]	0.341	[0.182 ; 0.490]

We recover three of the four regimes identified in Section 5.1.2 — *Ideal diversification* disappears with the elimination of one regime, which suggests an underlying consistency between the financial signal and the macroeconomic signal. The pure macroeconomic structure covers the expected episodes well: the *Financial crisis* (R0) captures the Dotcom, GFC, and COVID phases; the *Normal* regime (R1) covers the *Great Moderation* and the post-GFC *low-vol era*; *Inflationary stress* (R2) identifies the stagflation of the 1970s and the post-COVID inflationary cycle of 2022–2024.

5.2.2 Graphical analysis of the results obtained

To check the economic consistency of the identified regimes, we overlay the sequence of states decoded by Viterbi on the S&P500 wealth index (Figure 4) and on the implied S&P500/10-year T-bond correlation (Figure 5).

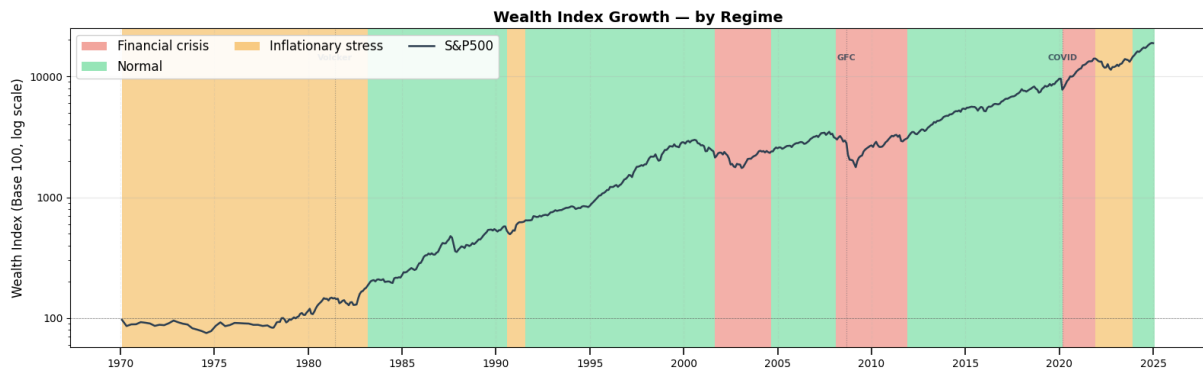


Figure 4: S&P500 wealth index (base 100, logarithmic scale) and decoded HMM regimes.

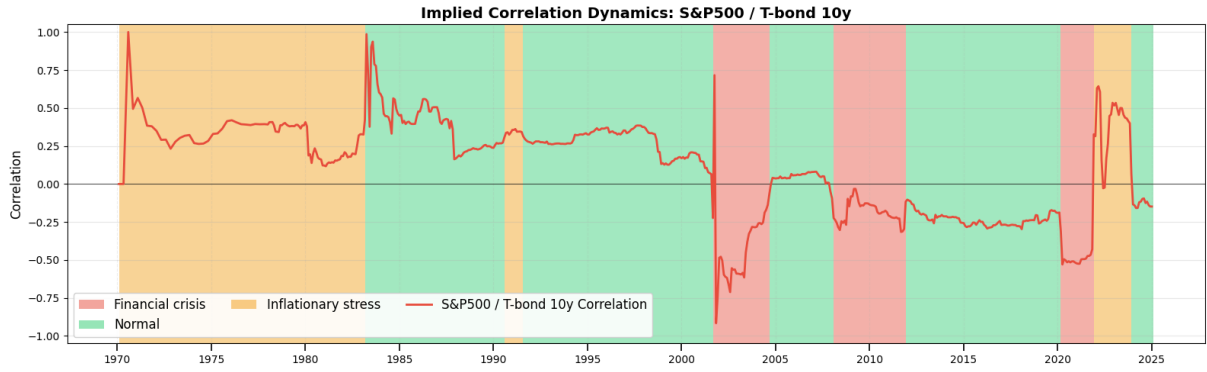


Figure 5: Implied conditional S&P500 / 10-year T-bond correlation and decoded HMM regimes.

The *inflationary stress* bands (R2) precisely frame the three major episodes of price tension over the period: the stagflation of 1970–1982 marked by the two oil shocks and the Volcker tightening, the pre-recession inflation of 1990–1991 linked to the Gulf War, and the post-COVID cycle of 2022–2024 with an inflation peak at 9.1% and the fastest Fed hiking cycle since Volcker. Over each of these windows, the implied S&P/T-bond correlation remains positive, which confirms the signature of a regime in which the rate factor dominates the variance of both asset classes.

The *financial crisis* bands (R0) clearly identify the Dotcom crash and the 2001 recession, the 2008 global financial crisis prolonged by the European sovereign crisis until 2012, as well as the COVID shock of March 2020. The S&P/T-bond correlation then swings into negative territory (down to -0.5), the classic signature of a *flight-to-quality*. The remaining phases, identified as *normal* (R1), coincide with the recognised periods of low systemic volatility: the end of the Volcker disinflation (1984–1989), the bull market of the 1990s, and the post-GFC *low-volatility era* (2013–2019).

5.2.3 The limitations of this approach

The identification of purely macroeconomic regimes makes it possible to answer the initial question in part: these regimes do coincide with financial phases that are distinct in volatility and correlation, and the endogenous swing of the S&P/T-bond correlation around the year 2000 is recovered without calendar information, which constitutes a *quasi out-of-sample* validation of the economic interpretation.

The comparison of the robustness scores with the market HMM (detailed in Appendix C.11) reveals, however, a trade-off. The pure macroeconomic HMM displays markedly higher persistence ($\mathbb{E}[D]_{\min} = 37.4$ months versus 6.0) and a higher decoding conviction (98.4 % versus 81.2 %), reflecting the structural nature of its regimes. In return, its separability falls (66.7 % versus 81.3 %): with only three regimes, certain volatility pairs no longer differ significantly, notably on the T-Bond between Crisis and Normal. Overall, the global score remains slightly higher (83.1 % versus 80.56 %), but at the cost of lower financial granularity.

Three limitations follow from this. First, short stress episodes (Black Monday 1987, LTCM 1998, regional banks 2023) remain invisible: their duration of less than two months is incompatible with the strong persistence of the model ($P(\text{stay}) > 0.97$). Second, the FRED indicators are released with a lag of one to several months, hardly compatible with the kinetics of financial crises that play out in a few days. Finally, the macro-financial correspondence is not bijective: certain financial tensions emerge in a favourable macroeconomic context (the 1997–1998 Asian crisis), and vice versa (the 1987 crash).

It is precisely this trade-off between macroeconomic readability and financial granularity that motivates the hybrid approach developed in the following section: keeping regimes defined on financial returns — hence reactive and finely separated — while letting the macroeconomy drive their transitions.

5.3 Latent market regimes conditioned on the macroeconomy

The identification of market regimes appears to constitute an essential first step in understanding risk dynamics. However, such an approach (Section 5.1) remains purely descriptive and does not allow us to understand the mechanisms underlying the transitions between regimes.

Yet it is reasonable to assume that the changes in these regimes depend on the economic environment. For example, a widening of credit spreads or an inversion of the yield curve may signal a deterioration in financial conditions and increase the probability of moving towards a stress regime. In order to incorporate this information, we use the HMM-TVTP model to model market regimes whose transitions are impacted by the macroeconomy. The observations are $Y_t = (r_t^{S\&P}, r_t^{T-Bond\ 10y})$ and the covariates $X_t = (\text{Inflation YoY}_t, \text{Credit-Spread Diff}_t, \mathbf{1}_{Slope_t \leq 0})$ (chosen on the basis of the results of the correlation matrices and the PCA in Appendix B.2 and B.1).

To ensure the comparability of the results with Section 5.1, we keep the same number of latent states $K = 4$.

5.3.1 Characterisation of the regimes (emission parameters)

The HMM-TVTP model allows us to obtain the following per-regime covariance matrices:

Table 6: Characteristics of the TVTP market regimes (annualised volatilities in % and 95% Bootstrap CI)

Regime	Freq.	Duration	$\sigma_{S\&P}$	CI _{95%}	σ_{TBond}	CI _{95%}	Correlation	CI _{95%}
0	0.192	8.27	21.09	[17.22 ; 24.80]	6.38	[5.39 ; 7.33]	-0.443	[-0.641 ; -0.221]
1	0.576	31.02	9.43	[8.69 ; 10.17]	6.14	[5.64 ; 6.60]	0.393	[0.316 ; 0.467]
2	0.098	7.48	16.16	[12.82 ; 19.02]	14.05	[11.20 ; 16.47]	0.320	[0.074 ; 0.530]
3	0.135	6.47	6.84	[5.98 ; 7.62]	4.77	[4.15 ; 5.32]	-0.201	[-0.362 ; -0.030]

Thus, we can note that we recover market regimes similar to those obtained with the first model, which seems to confirm a stability of these regimes. As the analysis of the latter has already been carried out in Section 5.1.2, we will focus more on the macroeconomic transitions.

5.3.2 Dynamics of the regimes (transition parameters)

Table 7: HMM-TVTP transition parameters: intercepts and macroeconomic sensitivities

Regime	Intercepts β_{ij}				Macroeconomic weights ω_j		
	R0	R1	R2	R3	Inflation	Credit spread	$\mathbf{1}_{Slope \leq 0}$
R0	2.986	-0.247	-2.725	-0.015	-0.046	-0.042	0.194
R1	-0.569	4.316	-2.321	-1.426	0.237	0.144	-0.785
R2	-0.078	1.000	3.566	-4.487	0.773	0.608	0.126
R3	1.082	-3.800	-0.733	3.451	-0.964	-0.710	0.465

The coefficients in Table 7 thus make it possible to distinguish a structural component (intercepts β_{ij}) and a cyclical component (sensitivities ω_j).

First, the diagonal intercepts remain largely dominant and positive for regimes 0, 1, 2, and 3, which confirms a strong intrinsic persistence of the states, already observed in the homogeneous model. Regime 1 remains the most stable, with a high own intercept ($\beta_{11} = 4.316$), which suggests that in the absence of a macroeconomic shock, the economy tends naturally to converge towards this central normalised-market regime. Regimes 2 and 3 also display significant inertia ($\beta_{22} = 3.566$, $\beta_{33} = 3.451$ and cf. Appendix C.6), while regime 0 appears relatively less stable.

Second, the macroeconomic variables modify the probability of entering each regime in an economically coherent manner. A rise in inflation increases the relative probability of regime 2 ($\omega_\pi = 0.773$), identified as an inflationary stress regime characterised by high simultaneous volatility of equities and bonds. Likewise, a widening of the *credit spread* also increases the probability of this regime (0.608), which reflects the sensitivity of this state to financial tensions and to the tightening of credit conditions. Conversely, the coefficients associated with regime 3 are strongly negative for inflation (-0.964) and the *credit spread* (-0.710). This means that a *fall* in inflation or in the credit spread increases the probability of entering this regime, which characterises a sharp disinflation episode or a growth shock more than a classic inflationary crisis. This result is consistent with the interpretation of regime 3 as a *flight-to-quality* episode, marked by a flight towards sovereign bonds.

Finally, the inversion of the yield curve plays an important discriminating role. Its negative coefficient on regime 1 (-0.785) indicates that a *yield curve inversion* reduces the probability of remaining in the normal regime. Symmetrically, its positive coefficients on regimes 0 and 3 suggest that an extreme flattening or an inversion of the curve more often precedes phases of transition or marked slowdown. The detail of the significance levels is in Appendix C.6 and C.7 (here we present the raw results of the macroeconomic weights to facilitate interpretation).

5.3.3 Temporal evolution of the regimes and the correlation

Figures 6 and 7 respectively represent the evolution of the cumulative wealth index, segmented by the HMM-TVTP regimes, and the correlation between the S&P and the T-Bond. Once again, in descriptive mode, we recover results close to those of Section 5.1.4, which analyses the results.

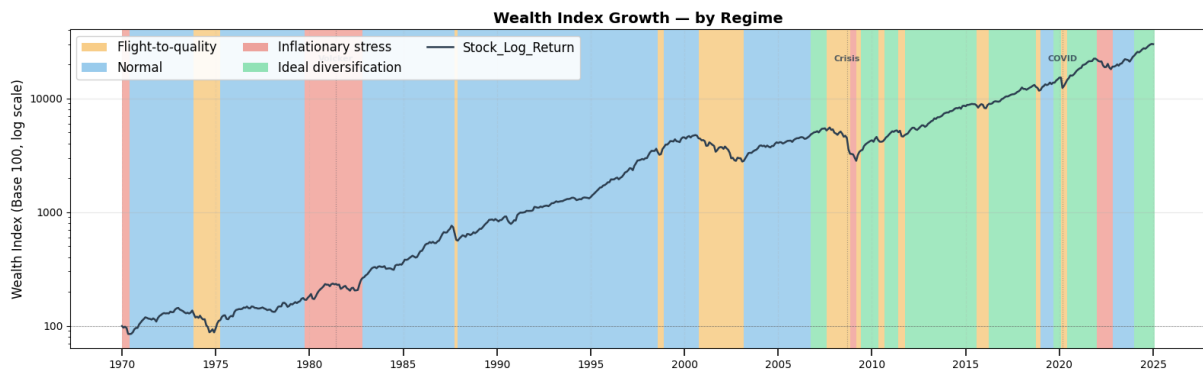


Figure 6: S&P - Cumulative return by HMM-TVTP market regime

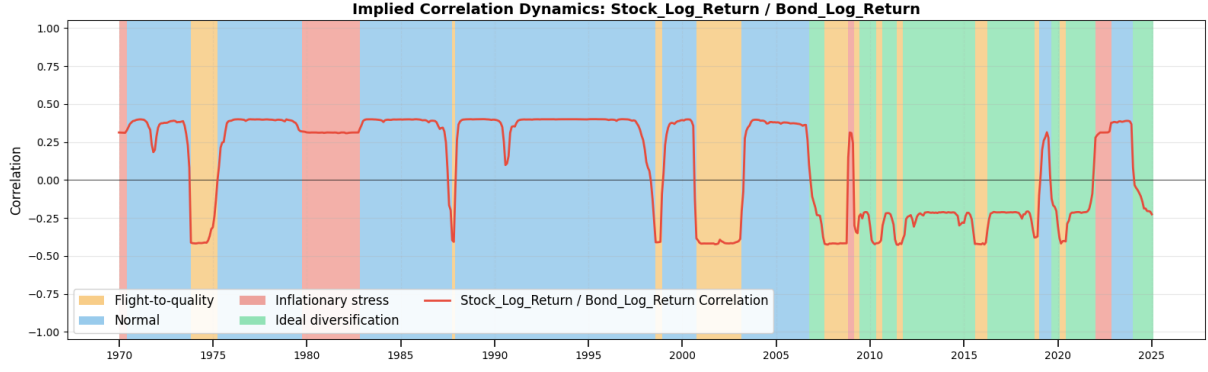


Figure 7: Conditional correlation between the S&P and the 10-year T-Bond by HMM-TVTP regime

6 Dynamic allocation

6.1 Backtest methodology

The allocation is evaluated on an *expanding window* to exclude any look-ahead bias: at each date t , the model is re-estimated on the history 1974– t and the resulting weights are applied to the returns of the period $t + 1$. Starting in 1974 excludes the Bretton Woods transition (1971–1973), and the backtest begins in 1995 to ensure an out-of-sample evaluation over a period of modern markets.

To limit convergence towards local optima, we alternate between *warm-start* (initialisation with the parameters of the previous model, which ensures numerical stability and semantic consistency of the regimes) and full periodic re-initialisation (*k-means* or random draw). Stability is monitored at each iteration by the global score defined in Appendix A.2. The model used is the HMM-TVTP with observations $(r_t^{S\&P}, r_t^{T-Bond\ 10y})$ and covariates inflation (differenced over 12 months) and the credit spread (second difference), in order to capture the short-term dynamics of credit conditions and to make the model sensitive to rapid variations in financial tensions, interpreted as the acceleration of market stress, i.e. the first-order derivative of its variation, which is particularly informative for the detection of regime transitions.

6.2 Regime-conditional allocation

The architecture of the allocator follows a *Mixture of Experts* logic: each regime k is treated as an expert proposing a specific allocation $w_t^{(k)}$, and the final portfolio allocation W_t is obtained as a weighted average by the state probabilities $p_{k,t}$:

$$W_t = \sum_{k=1}^K p_{k,t} \cdot w_t^{(k)}$$

Each vector $w_t^{(k)}$ is determined by *Risk-Budgeting* (Appendix D.1), a method that takes as input the risk budgets associated with each asset. The performances include transaction costs of 10 bps. The maximum leverage is set to 1, excluding any recourse to borrowing.

Dynamic risk budget. Rather than imposing a fixed allocation, the risk budget allocated to the S&P $b_{S\&P}^{(k)}$ varies with the perceived risk of the regime, measured by the annualised volatility that a reference

60/40 portfolio would have in regime k :

$$Risk_k = \frac{\sigma_{60/40}^{(k)} - \sigma_{60/40,min}}{\sigma_{60/40,max} - \sigma_{60/40,min}}, \quad b_{S\&P}^{(k)} = b_{max} - Risk_k \cdot (b_{max} - b_{min})$$

where b_{min} and b_{max} are set by the user. This rule reduces exposure to risky assets in unstable regimes while maximising the capture of the risk premium in calm regimes. Two *overlays* complement this mechanism in order to make the best use of the regime structure.

Overlay 1 – Vol-Targeting. Each expert adjusts its exposure so as not to exceed a target volatility σ_{target} . The expected volatility of expert k is written $\sigma_t^{(k)} = \sqrt{w^{(k)'} \Sigma^{(k)} w^{(k)}}$ with $w^{(k)} = [w_{S\&P}^{(k)}, w_{T10}^{(k)}]'$, and its gross weight is adjusted by a leverage factor:

$$L_t^{(k)} = \min \left(\frac{\sigma_{target}}{\sigma_t^{(k)}}, 1 \right)$$

Overlay 2 – Stress and loss of diversification. This overlay deals specifically with regimes where high volatility and positive correlations coexist: diversification benefits disappear there since no asset protects the other. The stress regime k^* is defined by

$$k^* = \arg \max_k (\mathbf{1}_{\rho_k > 0} \cdot Risk_k),$$

and only this regime is subject to a progressive reduction of the maximum authorised leverage:

$$L_{cap}^{(k)} = \begin{cases} 1 - Risk_k \cdot (1 - L_{min}) & \text{if } k = k^* \\ 1 & \text{otherwise} \end{cases}$$

The residual cash (weight $1 - \sum_k w^{(k)}$) is remunerated at the risk-free rate approximated by the *TB3MS yield* from FRED (Federal Reserve Bank of St. Louis, 2026).

6.3 Prediction of the volatilities and the correlation between the S&P and the 10-year T-Bond

The figure in Appendix D.1 represents the predicted volatilities of the S&P and the T-Bond as well as their predicted correlation. Thus, for the 2001-2002 crisis, we seem to observe a rise in volatility from the start of 2001 that will intensify until around 2003, which seems consistent with 1. Moreover, we seem to observe a genuine prediction in terms of stress for 2008, with volatility rising at the end of 2007 / beginning of 2008. For March 2020 (COVID-19), the model is only reactive, which is, *a priori*, consistent given that this shock was exogenous. Finally, for 2022, equity volatility seems to rise gently before the April/May shock.

Furthermore, in terms of correlation, we do observe the shift towards rather negative-correlation regimes after 2000 with the first crisis of 2001-2003, then 2008, followed by calm periods as mentioned in Section 5.1.4.

6.4 Performance

Table 8 compares several performance metrics (whose definitions are in Appendix D.2) between our TVTP HMMs and benchmarks (60 % S&P and 40 % T-Bond, and *Risk Parity* – a *Risk Budgeting* in Appendix D.1 with the same budget for each asset – with rolling covariance matrices). Thus, we find that the second *benchmark*, the *Rolling Risk Parity*, already improves on the 60/40 in terms of risk control but loses in return. Next, we can note that our 100 % exposed HMM-TVTP already significantly improves the metrics. Indeed, the return is higher for a lower risk, which gives us a Sharpe ratio of 0.97. Moreover, the *maximum drawdown* is significantly reduced compared to the 60/40 benchmark by nearly 10 % and by 2 % relative to the *Rolling Risk Parity*. Finally, the addition of the *overlays* makes it possible to reduce the *maximum drawdown* by 6 % while reducing volatility and increasing return, which proves, *a posteriori*, that these *overlays* make it possible to better exploit the regime structure of the model. The final model achieves a Sharpe ratio of 1.08, which means that for 1 % of risk, our model remunerates us with 1.08 % of excess return (relative to the risk-free rate), which appears to be a good performance over the 1995–2025 period. The performance and weight-attribution charts are respectively in Figure 8 and Figure 9.

Table 8: Comparison of historical performances (1995-2025)

Indicator	HMM-TVTP (Overlay 7%)	HMM-TVTP Pure (100% Expo)	Rolling Risk Parity	Benchmark 60/40
Ann. Return	9.26%	9.10%	7.04%	8.68%
Ann. Volatility	6.22%	6.82%	5.64%	7.67%
Net Sharpe ratio	1.08	0.97	0.82	0.81
Max Drawdown	-11.04%	-17.00%	-19.13%	-27.00%
Calmar ratio	0.83	0.54	0.37	0.32

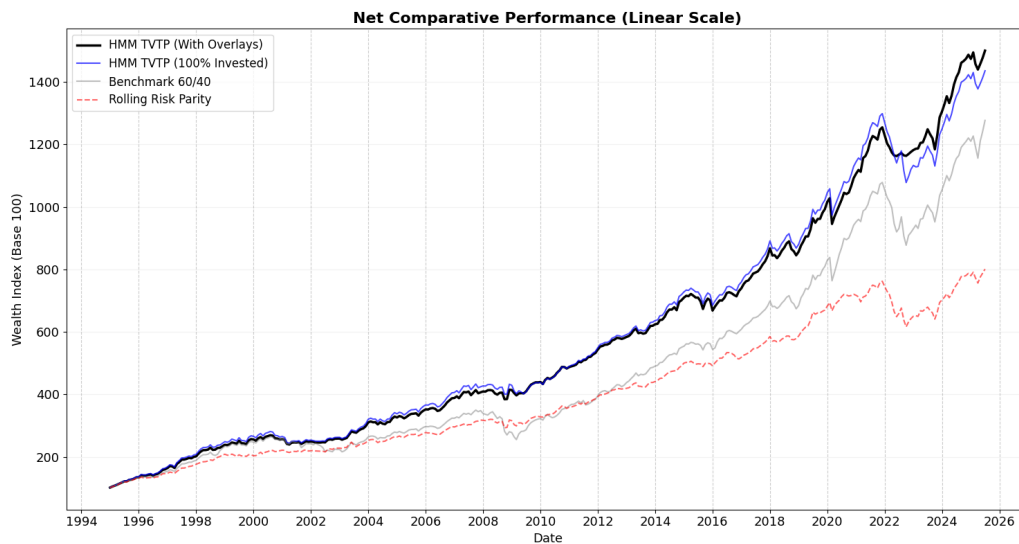


Figure 8: Cost-net performances (10bps per transaction) of our model and the benchmarks

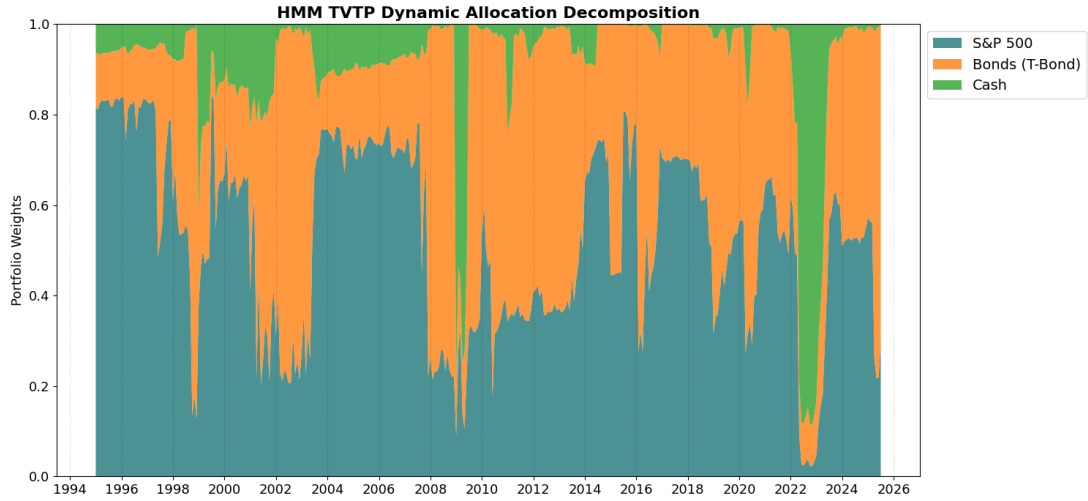


Figure 9: Attribution of the weights on the assets by the HMM-TVTP

6.5 Case study number 1: the 2008 Financial Crisis

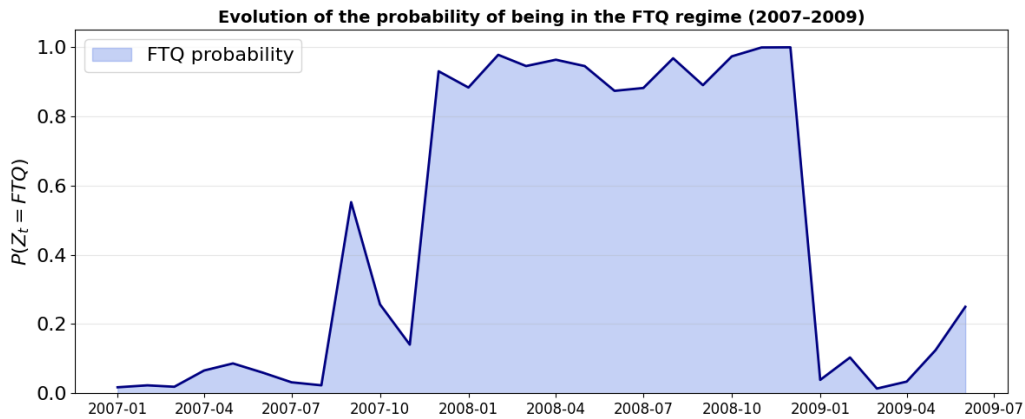


Figure 10: Evolution of the probability of being in an FTQ-type regime in 2008

The analysis of the 2007–2009 period (Figure 10) illustrates the ability of the HMM-TVTP model to identify structural breaks in real time. The detected regime corresponds to the theoretical characteristics of the *Flight-to-Quality* (FTQ): high equity volatility, moderate bond volatility, and a negative S&P 500/T-Bond correlation (in January 2008: 18.8 %, 6.2 % and -0.39 respectively).

As early as the second half of 2007 – while the stock indices are still close to their peaks –, the probability of the FTQ regime rises from less than 10 % to more than 50 % in September 2007, that is, from the first signs of the freezing of the interbank market (BNP Paribas, August 2007). The model therefore detects the change of regime before the acceleration of the fall of the S&P 500. Throughout 2008, this probability remains anchored above 80 %. The negative correlation here preserves the protective role of bonds, justifying the maintenance of a 100 % exposure (Figure 9) in view of a potential rebound. At the end of 2008, the model switches towards a pure crisis regime (January 2009: $\sigma_{S\&P} = 25.5$ %, $\sigma_{TBond} = 15.1$ %, $\rho = 0.32$, probability 85.6 %).

The macroeconomic impact breaks down into two distinct phases. In the early detection phase (2007 – early 2008), neither the credit spread nor inflation provides an exploitable signal: the switch towards

the FTQ is driven by the returns alone. In the confirmation phase (November 2008), the explosion of the credit spread ($+6\sigma$ on $credit_spread_{diff}$) reinforces the verdict of the model, then orienting almost all the transitions towards the FTQ – including from the normal regime (Figure 11).

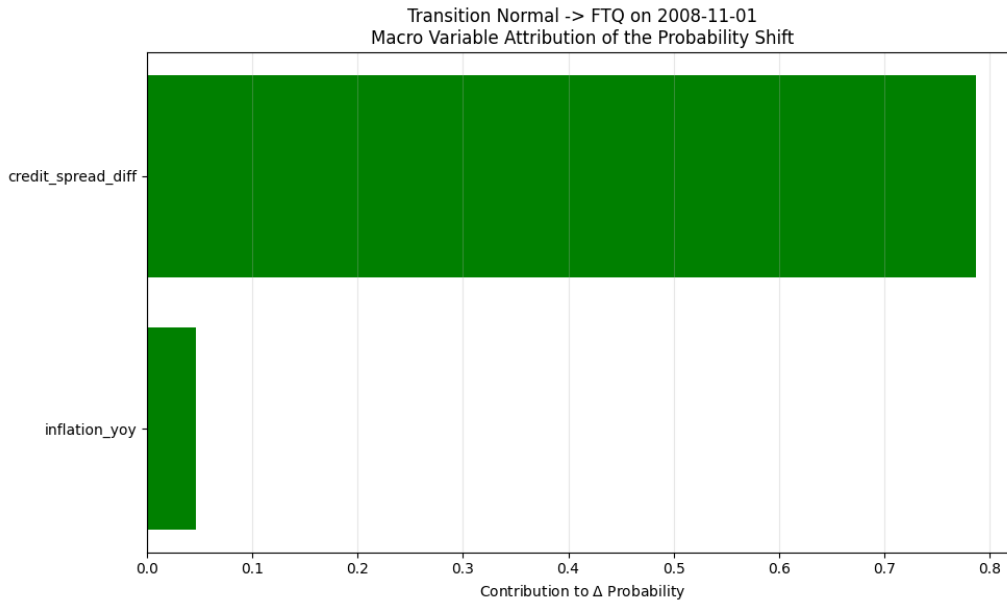


Figure 11: Attribution of the evolution of the transition matrix in November 2008 (computed by *Integrated Gradients*, cf. Appendix D.3): +78.10% of total variation, of which 73.04% due to the credit spread

6.6 Case study number 2: the 2022 Inflationary Crisis

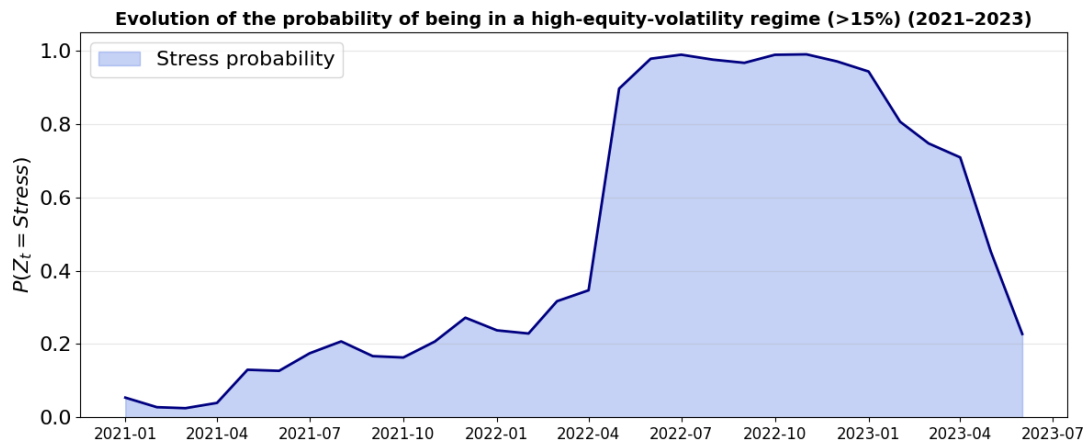


Figure 12: Evolution of the probability of being in a high equity volatility regime ($> 15\%$) in 2022

To analyse 2022, we reason rather in terms of "positive-correlation regimes" or "high-volatility regimes" rather than a precise regime, because the model detected signals as early as the end of 2021 / beginning of 2022, switching clearly (with probability 91.2 %) towards the classic stagflation/systemic crisis regime (21.3 % equity volatility, 12.3 % bond volatility and 0.12 correlation) in May 2022.

Thus, the observation of the regime probabilities reveals a certain asymmetry between the detection of volatility and that of correlation:

- **The correlation signal as a state indicator:** As early as January 2022, with only the returns and the macroeconomy of 2021 as observations, the probability of the positive-correlation regime undergoes a near-instantaneous phase transition, rising from 10% to 80% (13).
- **The volatility signal as a momentum indicator:** Although the probabilities of the high-volatility regimes display an upward slope from the end of 2021, which nonetheless raises the overall risk (since these regimes are very risky), the real break is in May 2022.

Indeed, we can observe in Figure 9 that the model reduces its exposure from the start of 2022 because the probability of being in the regime defined as a systemic crisis gradually increases, which makes it possible to reduce the decline.

Regarding the macroeconomic impact, we can take the example of the transition from January 2022 to February 2022 from the calm regime (7.4% equity volatility, 5.3% bond volatility and a correlation of 0.06) to the stagflation regime (21.2% equity volatility, 12.3% bond volatility and a correlation of 0.12). Indeed, we note in Figure 14 that inflation is the main cause.

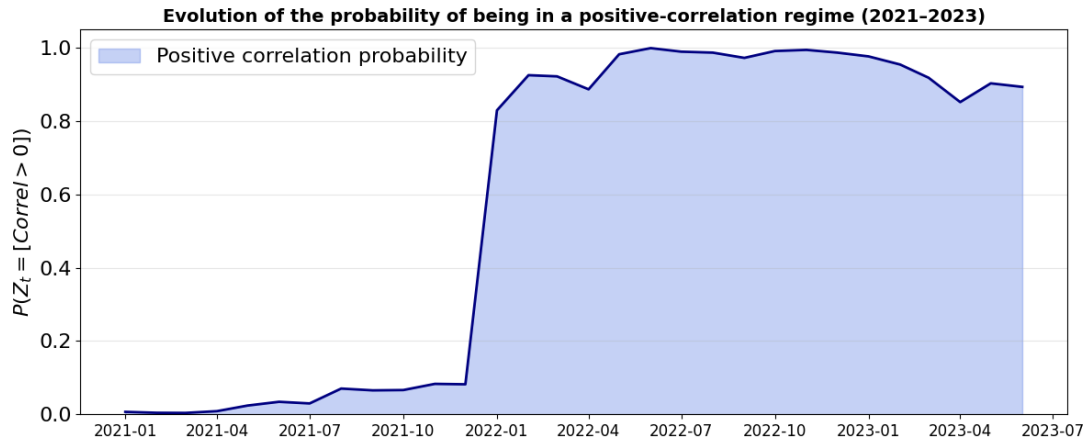


Figure 13: Evolution of the probability of being in a positive-correlation regime in 2022

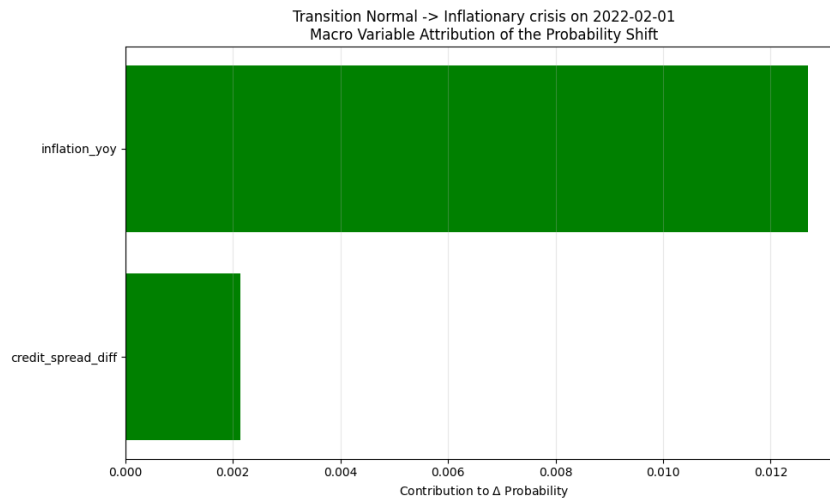


Figure 14: Attribution of the evolution of the transition matrix in February 2022 (computed by *Integrated Gradients*, cf. Appendix D.3): +1.71%, of which 1.52% due to inflation

7 Conclusion

This work aimed to determine whether it was possible to identify statistically robust latent market regimes whose transitions are driven by the macroeconomy, and to derive from them a dynamic allocation outperforming static or rolling benchmarks over a long cycle. The approach deployed combines a probabilistic latent-regime modelling (HMM and HMM-TVTP) with a framework of conditional allocation of the *Mixture of Experts* type, evaluated on an *expanding window* over the 1995–2025 period. The results obtained make it possible to answer the three components of this problem positively.

First, the existence of statistically robust regimes is validated by the convergence of the three descriptive approaches. The homogeneous HMM fitted on the S&P 500 and the 10Y T-Bond identifies four economically interpretable regimes — *ideal diversification*, *normal*, *inflationary stress* and *flight-to-quality* —, whose robustness is confirmed by a global score $s_G = 80.6\%$ aggregating separability, numerical stability, persistence and detection conviction. The pure macroeconomic HMM moreover recovers, without calendar information, the major financial episodes (stagflation, Dotcom, GFC, COVID, 2022–2024 inflationary cycle), which constitutes a validation through informational independence: the macroeconomic signature of the financial regimes is so robust that it emerges from purely fundamental data.

Second, the HMM-TVTP demonstrates that the transitions between regimes are indeed driven by the macroeconomy. The estimated sensitivities are economically coherent: inflation and the widening of the *credit spread* increase the probability of the inflationary stress regime, while the inversion of the yield curve signals the switches towards the defensive regimes. The case studies confirm this mechanism in real time: in 2008, the model detects the switch towards the *flight-to-quality* as early as the second half of 2007, the explosion of the *credit spread* in November reinforcing the verdict with an attributed contribution of 73 % by *Integrated Gradients*; in 2022, it is on the contrary inflation that constitutes the main driver of the switch towards stagflation, the model reducing its exposure from the start of the year.

Third, the regime-conditional allocation clearly outperforms the standard benchmarks. The HMM-TVTP portfolio enriched with the *vol-targeting* and diversification-stress management overlays reaches a net Sharpe ratio of 1.08 against 0.81 for the 60/40 and 0.82 for the *Rolling Risk Parity*, while reducing the *maximum drawdown* by nearly 16 points. This improvement does not reflect a mere compression of risk but a better exploitation of the conditional structure of the regimes, the annualised return remaining higher than that of the benchmarks.

The main contribution of this work therefore lies in the articulation of an economic reading of the market phases and an operational allocation framework: the HMM-TVTP provides both an interpretable grid of risk dynamics and an exploitable probabilistic signal, restoring endogenously the major historical breaks in the equity–bond correlation.

Several limitations must nonetheless be recalled. First, the conditional normality assumption, retained for reasons of analytical tractability, underestimates the thickness of the distribution tails, while the strong persistence of the regimes makes it difficult to detect very short stress episodes (Black Monday 1987, LTCM 1998, regional banks 2023). Second, the monthly frequency smooths the variance and prevents the capture of intra-month high-volatility episodes, which could have been addressed by an HMM-GARCH-type modelling at higher frequencies. Third, the first-order Markovian structure implicitly imposes a geometric law on the duration of the regimes: the future state depends only on the present state (and possibly on covariates); whereas an approach using hidden semi-Markov models (HSMM) would make

it possible to better capture the long memory of economic cycles. Finally, the publication lag of the macroeconomic indicators may limit the responsiveness of the signal in real time: it would therefore be relevant to test, in future work, market covariates available in real time (VIX, MOVE, sovereign spreads).

Ultimately, this work shows that the combination of a probabilistic latent-regime modelling and a conditional steering by the macroeconomy makes it possible to build an allocation that is both interpretable and high-performing. Beyond the improvement of the risk-return metrics, the proposed framework offers an economic reading grid of the market phases that restores, endogenously, the major historical breaks in the equity–bond correlation.

A Supplementary material on hidden Markov models and their validation

A.1 EM estimation and Viterbi decoding

The complete log-likelihood of the HMM over a sequence $O_{1:T} = (O_1, \dots, O_T)$ is written

$$\ell(\theta) = \log \mathbb{P}(O_{1:T} \mid \theta) = \log \sum_{z_1, \dots, z_T} \mathbb{P}(O_{1:T}, z_{1:T} \mid \theta),$$

a sum that admits no closed form because of the marginalisation over the latent states. The Baum-Welch algorithm is a specialisation of EM to latent-variable models: the E-step computes the *a posteriori* probabilities of the hidden states via the Forward-Backward algorithm, and the M-step updates θ by maximising the expectation of the complete log-likelihood under the current *a posteriori* distribution. In the HMM-TVTP case, the update of the coefficients (β_{ij}, ω_j) in the M-step no longer admits an explicit solution and is solved numerically by L-BFGS.

For the initialisation of the algorithm on the emission parameters, we use, at our discretion, a random initialisation with *warm restart*, a K-means clustering on the observations, or, for the TVTP, a prior homogeneous HMM fit. For the transitions of the HMM-TVTP, we use a logistic regression between the macroeconomic variables and the regimes obtained by a homogeneous HMM. The Viterbi algorithm then makes it possible to infer the most likely sequence of states by dynamic programming.

A.2 Validation tests for the probabilistic models

The validation of an HMM fit does not reduce to the log-likelihood: for a model to be usable in asset allocation, its regimes must be statistically distinct, numerically stable, economically persistent, and produce high-conviction signals. We formalise these four dimensions through four scores normalised in $[0, 1]$, aggregated into a global score s_G .

A.2.1 Separability score s_{SE}

Levene test. To compare the volatility between two regimes i and j on a given asset, σ_i and σ_j , we use the Levene test. The null hypothesis is:

$$H_0 : \sigma_i^2 = \sigma_j^2$$

The choice of Levene rather than Bartlett or F is motivated by its robustness to the non-normality of the conditional returns.

Fisher-Z test. To compare the correlation of two assets between two regimes, ρ_i and ρ_j , we use the Fisher transformation, which stabilises the variance of the estimator:

$$z = \frac{1}{2} \ln \frac{1+r}{1-r}$$

The null hypothesis is then:

$$H_0 : z_i = z_j$$

Bonferroni correction. With K regimes and D assets (or $\binom{D}{2}$ pairs), the number of pairwise tests grows quadratically and the cumulative type-I error rate explodes. The Bonferroni correction multiplies the

p-values by the number of comparisons, guaranteeing a FWER (*family-wise error rate*) controlled at α . This correction is conservative, but suited to an operational context where a false positive on separability can lead to superfluous transaction costs.

Bootstrap on the conditional statistics. The confidence intervals on the conditional volatilities and correlations by regime are obtained by non-parametric bootstrap (5 000 resamplings) on the observations assigned to each regime by Viterbi. This approach makes no parametric assumption on the conditional distribution.

We thus define:

$$s_{SE} = \frac{n_{\text{significant}}}{n_{\text{total}}}.$$

A score close to 1 indicates a clear segmentation; a low score signals a blurred model, risky for a *volatility targeting* strategy.

Critique of the statistical tests:

These results must be qualified, as certain assumptions of the tests are partially violated:

- **Independence (i.i.d.):** The series exhibit autocorrelation, but conditionally on a latent state, the residuals become approximately i.i.d. within each regime.
- **Normality:** The Fisher test assumes Gaussian distributions and may over-reject the null hypothesis in the presence of fat tails.
- **Assignment:** To assign an observation to a regime, we use the Viterbi algorithm and, in effect, do not take this classification uncertainty into account.

A.2.2 Numerical stability score s_{ST}

To analyse the statistical coherence of the transition parameters θ_T , a local analysis of the likelihood surface and of the persistence of the regimes is carried out. To estimate the parameters, we seek to maximise the log-likelihood $l(\theta)$. Thus, if θ^* is the optimal parameter vector, then the Hessian of the log-likelihood $\nabla^2 l(\theta^*) := \mathcal{H}(\theta^*)$ must be a negative-definite matrix. In other words, all its eigenvalues $\lambda \in \text{Sp}(\mathcal{H}(\theta^*))$ must be strictly negative.

Since we are only interested in θ_T , we denote hereafter by $\mathcal{H}_T(\theta_T^*)$ the sub-matrix. In the statistical framework, we are interested rather in the Fisher information $\mathcal{I}_T(\theta_T^*) = -\mathcal{H}_T(\theta_T^*)$, which must therefore have strictly positive eigenvalues.

Moreover, we are also interested in the *condition number* of $\mathcal{I}_T(\theta_T^*)$, defined by:

$$\kappa(\mathcal{I}_T(\theta_T^*)) := \kappa^* = \frac{\lambda_{\max}}{\lambda_{\min}}$$

where $\lambda_{\max}$ and $\lambda_{\min}$ denote respectively the largest and the smallest eigenvalue of $\mathcal{H}_T(\theta_T^*)$.

A high *condition number* indicates a near-singular matrix, which translates into instability of the parameter estimation and a strong sensitivity to sample fluctuations. Conversely, if it is low, this suggests a more stable and better-identified estimation.

Thus, we define the stability score by:

$$s_{ST} = (1 - 0.9 \mathbf{1}_{\{\lambda_{\max} \geq 0\}}) \cdot \max\left(0, 1 - \frac{\kappa^*}{12}\right)$$

A.2.3 Significance of the transition parameters

The estimation of the uncertainty on the transition parameters θ_T requires resolving a structural indeterminacy: for a given originating regime i , the sum of the transition probabilities must satisfy $\sum_{j=1}^K p_{ij} = 1$. This summation constraint induces a perfect collinearity that would make the Fisher information matrix $\mathcal{I}_T$ singular.

To ensure the identification of the model, we adopt a relative *log-odds* approach by fixing a reference regime r_{ref} . The estimated parameters (intercepts β_{ij} and covariate weights ω_j) for a transition from i towards a regime j are then defined relative to this reference regime r_{ref} :

$$\eta_{ij}(X_t) = \ln\left(\frac{p_{ij}(X_t)}{p_{i,r_{ref}}(X_t)}\right) = b_{ij} + w_j^\top X_t, \quad \forall j \neq r_{ref}$$

with $\eta_{i,\text{ref}} = 0$ by construction, $b_{ij} = \beta_{ij} - \beta_{i,r_{ref}}$ and $w_j = \omega_j - \omega_{r_{ref}}$. The reduced parameter vector $\theta_{T,\text{red}}$ thus defined makes it possible to obtain a full-rank Hessian $\mathcal{H}_T$.

The standard errors of the parameters are then extracted from the diagonal of the asymptotic variance-covariance matrix, obtained by inverting the observed Fisher information at the optimal point:

$$SE(\hat{\theta}_j) = \sqrt{[\mathcal{I}_T(\theta^*)^{-1}]_{jj}}$$

The statistical significance is then evaluated by a Wald test, where the test statistic $z = \hat{\theta}_j / SE(\hat{\theta}_j)$ asymptotically follows a standard normal distribution $\mathcal{N}(0, 1)$, allowing the computation of the p -values presented in the results.

A.2.4 Persistence score s_P

Homogeneous case (HMM) : for a homogeneous Markov chain, conditionally on being in a regime i at date t , the probability of remaining in this same regime at the following period is given by p_{ii} . Thus, the average duration of this regime follows a geometric law $\overline{D}_i \sim \mathcal{G}(1 - p_{ii})$. In effect, the average time spent in regime i is given by:

$$\mathbb{E}[\overline{D}_i] = \frac{1}{1 - p_{ii}}.$$

This quantity provides a direct measure of the persistence of the regimes: the closer p_{ii} is to 1, the more durable the regime.

Moreover, the stationary distribution $\pi = (\pi_1, \dots, \pi_K)$ of a Markov chain is defined as the solution of the system:

$$\pi^\top P = \pi^\top, \quad \sum_{i=1}^K \pi_i = 1 \tag{A.1}$$

In other words, once an eigenvector v associated with 1 has been found, it is normalised by $\pi_i = \frac{v_i}{\sum_k v_k}$ (note that the sum of the coordinates of v is necessarily different from 0, since by the Perron-Frobenius

theorem there exists an eigenvector whose coordinates are all positive and this space is of dimension 1, so the coordinates of v are of the same sign). This distribution represents the proportion of time spent in each regime in the long run, independently of the initial condition, provided that the chain is ergodic.

In addition, we can also estimate empirically the frequency of occurrence of the regimes from the estimated regimes $(\hat{z}_t)_{t=1}^T$, for example via the Viterbi algorithm:

$$\hat{f}_i = \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{\hat{z}_t=i\}}$$

Inhomogeneous case (HMM-TVTP) : for an inhomogeneous Markov chain, the average duration in regime i is therefore no longer geometric in the classical sense, but dependent on the trajectory of the covariates (X_t) . An approximation of the average conditional duration can be defined by:

$$\mathbb{E}[\overline{D}_i \mid X_1, \dots, X_T] \approx \frac{1}{T_i} \sum_{t=1}^T \frac{1}{1 - p_{ii}(X_t)} \mathbf{1}_{\{\hat{z}_t=i\}},$$

where $\hat{z}_t$ is the estimated state (for example via Viterbi).

Likewise, the stationary distribution no longer really makes sense when the transition matrix evolves over time. Thus, to obtain an average representation of the inhomogeneous chain, we propose to use the average transition matrix $\overline{P}_T = \frac{1}{T} \sum_t P_t$ in equation (A.1).

Finally, the frequency has exactly the same definition as in the homogeneous case.

Thus, we define the persistence score by:

$$s_P = \frac{1}{2} \left[\min\left(1, \frac{\overline{D}_{min}}{D_{seuil}}\right) + \min\left(1, \frac{\overline{f}_{min}}{f_{seuil}}\right) \right]$$

where $\overline{D}_{min}$, D_{seuil} , $\overline{f}_{min}$, f_{seuil} are respectively the minimum duration among the regimes, a minimum duration threshold, the minimum frequency among the regimes, and a minimum frequency threshold.

A.3 Conviction score s_{CI}

The filtered probability distribution $\gamma_{t,k} = \mathbb{P}(z_t = k \mid Y_{1:t}, \theta)$ quantifies the probability of the latent state belonging to each of the regimes conditionally on the observations available up to time t . However, these probabilities may be ambiguous (for example, two regimes with probability 50%). Thus, to quantify this certainty of the model, we use Shannon entropy.

Mathematical definition of the entropy Shannon entropy H_t measures the degree of disorder or uncertainty of a model. For hidden Markov models, it is defined by

$$H_t = - \sum_{k=1}^K \gamma_{t,k} \log \gamma_{t,k}$$

To facilitate interpretation, we normalise this measure into a confidence index $CI_t \in [0, 1]$:

$$CI_t = 1 - \frac{H_t}{\ln K}$$

Thus, CI_t is close to 1 when H_t is close to 0, that is, when the model is certain of the state ($\gamma_{t,k} = 1$ for some $k \in (1, \dots, K)$). Conversely, CI_t is close to 0 when H_t is close to $\ln(K)$, that is, when the model hesitates perfectly between all the states ($\gamma_{t,k} = \frac{1}{K}$ for all $k \in (1, \dots, K)$).

We then deduce an average confidence score:

$$s_{CI} = \frac{1}{T} \sum_{t=1}^T CI_t$$

A.4 Global score s_G

The four sub-scores are aggregated into a global robustness score:

$$s_G = w_{SE} \cdot s_{SE} + w_{ST} \cdot s_{ST} + w_P \cdot s_P + w_{CI} \cdot s_{CI}.$$

A model is retained only if each of the sub-scores exceeds a pre-established threshold; among the validated models, the configuration maximising s_{RB} is selected.

B Appendix to the section on empirical evidence of non-stationarity and financial regimes

B.1 Structure and correlation of the macroeconomic variables

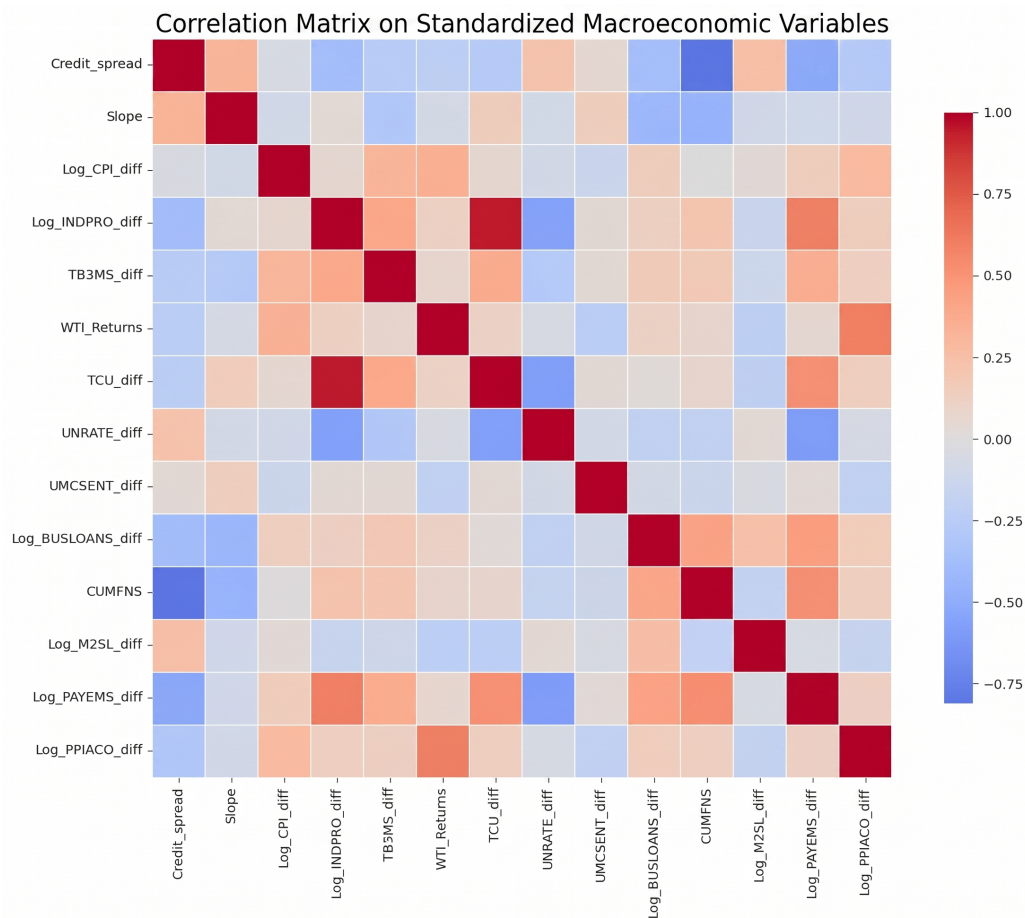


Figure B.1: Correlation matrix of the macroeconomic indicators (Inflation, Growth, Rates)

B.2 Factor extraction by Principal Component Analysis (PCA)

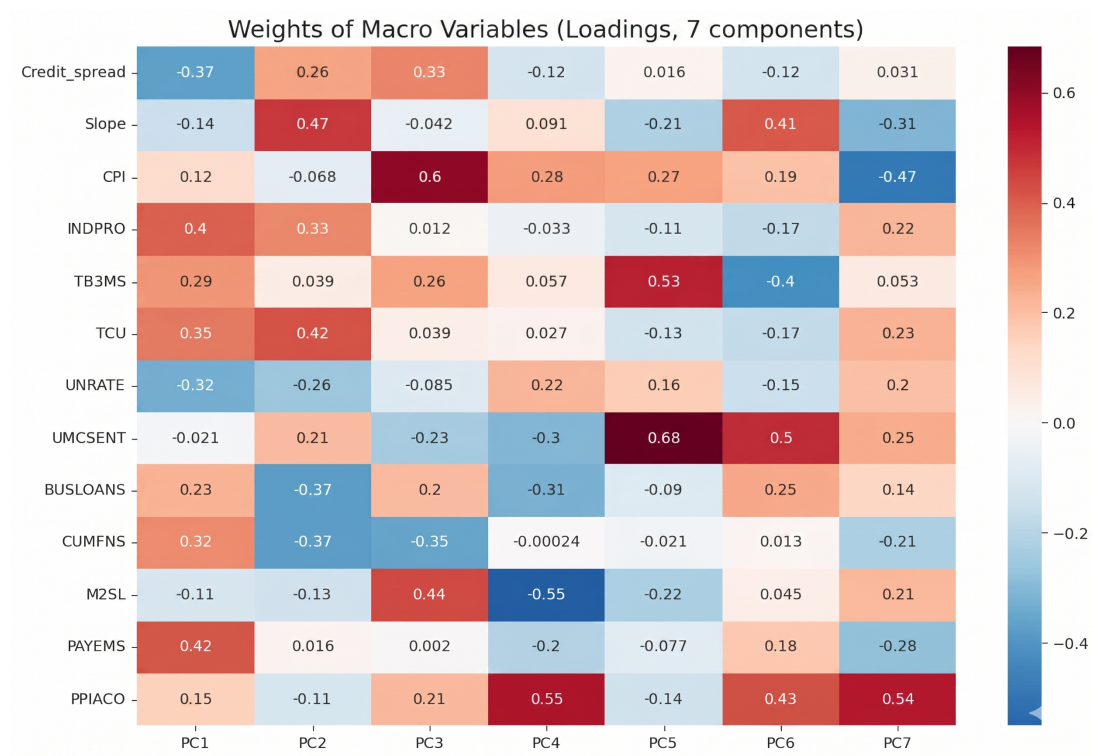


Figure B.2: Results of the Principal Component Analysis

C Appendices to the empirical results section

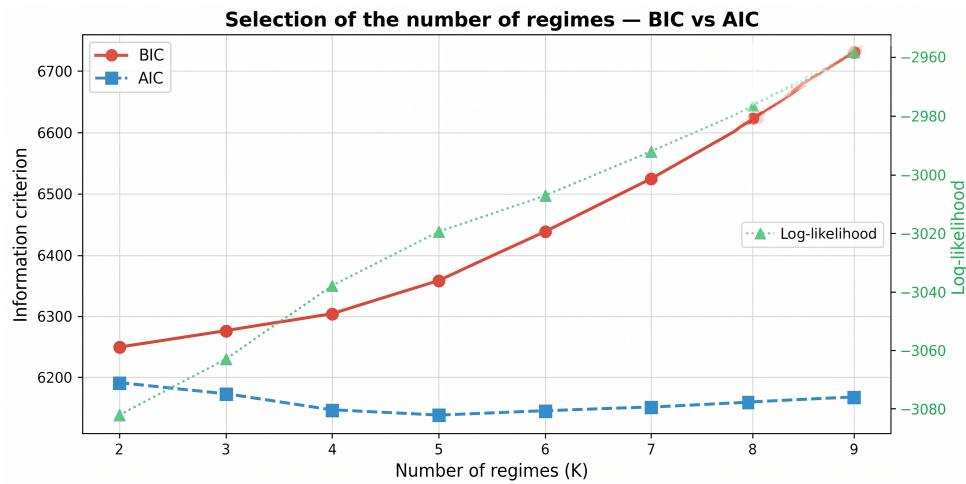


Figure C.1: Analysis of the optimal number of latent regimes according to AIC/BIC

C.1 Tests on the emission parameters

Table C.1: Separability tests of the market HMM regimes: volatility (Levene) and correlation (Fisher-Z)

Test	Regimes	Variable	Statistic	Adjusted p -value	Significant
Levene	0 vs 1	S&P	75.61	< 0.001	Yes
Levene	0 vs 3	S&P	80.29	< 0.001	Yes
Levene	1 vs 2	S&P	31.91	< 0.001	Yes
Levene	2 vs 3	S&P	53.88	< 0.001	Yes
Levene	1 vs 2	T-Bond	65.83	< 0.001	Yes
Levene	2 vs 3	T-Bond	64.93	< 0.001	Yes
Levene	0 vs 2	T-Bond	24.21	< 0.001	Yes
Levene	1 vs 3	S&P	17.74	< 0.001	Yes
Levene	1 vs 3	T-Bond	10.71	0.0137	Yes
Levene	0 vs 3	T-Bond	6.15	0.1657	No
Fisher-Z	0 vs 1	Corr (S&P, T-Bond)	-7.43	< 0.001	Yes
Fisher-Z	0 vs 2	Corr (S&P, T-Bond)	-4.69	< 0.001	Yes
Fisher-Z	1 vs 3	Corr (S&P, T-Bond)	5.81	< 0.001	Yes
Fisher-Z	2 vs 3	Corr (S&P, T-Bond)	2.93	0.0202	Yes
Fisher-Z	0 vs 3	Corr (S&P, T-Bond)	-2.56	0.0620	No
Fisher-Z	1 vs 2	Corr (S&P, T-Bond)	0.85	1.0000	No

Table C.2: Separability tests of the pure macroeconomic HMM regimes: volatility (Levene) and correlation (Fisher-Z)

Test	Regimes	Variable	Statistic	Adjusted p -value	Significant
Levene	0 vs 1	S&P500	24.56	< 0.001	Yes
Levene	1 vs 2	S&P500	20.67	< 0.001	Yes
Levene	1 vs 2	T-Bond 10y	15.73	0.001	Yes
Levene	0 vs 1	T-Bond 10y	3.10	0.473	No
Levene	0 vs 2	T-Bond 10y	1.88	1.000	No
Levene	0 vs 2	S&P500	0.92	1.000	No
Fisher-Z	0 vs 2	S&P500 / T-Bond 10y	-5.35	< 0.001	Yes
Fisher-Z	0 vs 1	S&P500 / T-Bond 10y	-3.83	0.001	Yes
Fisher-Z	1 vs 2	S&P500 / T-Bond 10y	-2.75	0.018	Yes

Table C.3: Separability tests of the HMM-TVTP regimes: volatility (Levene) and correlation (Fisher-Z)

Test	Regimes	Variable	Statistic	Adjusted p -value	Significant
Levene	0 vs 1	S&P	86.47	< 0.001	Yes
Levene	0 vs 3	S&P	77.55	< 0.001	Yes
Levene	1 vs 2	T-Bond	67.39	< 0.001	Yes
Levene	2 vs 3	T-Bond	59.33	< 0.001	Yes
Levene	2 vs 3	S&P	47.66	< 0.001	Yes
Levene	1 vs 2	S&P	28.08	< 0.001	Yes
Levene	0 vs 2	T-Bond	25.38	< 0.001	Yes
Levene	1 vs 3	S&P	17.10	0.0005	Yes
Levene	1 vs 3	T-Bond	9.69	0.0234	Yes
Levene	0 vs 3	T-Bond	7.03	0.1026	No
Levene	0 vs 2	S&P	3.00	1.0000	No
Levene	0 vs 1	T-Bond	0.06	1.0000	No
Fisher-Z	0 vs 1	Corr (S&P, T-Bond)	-7.58	< 0.001	Yes
Fisher-Z	1 vs 3	Corr (S&P, T-Bond)	6.23	< 0.001	Yes
Fisher-Z	0 vs 2	Corr (S&P, T-Bond)	-4.66	< 0.001	Yes
Fisher-Z	2 vs 3	Corr (S&P, T-Bond)	3.32	0.0055	Yes
Fisher-Z	0 vs 3	Corr (S&P, T-Bond)	-2.02	0.2618	No
Fisher-Z	1 vs 2	Corr (S&P, T-Bond)	0.57	1.0000	No

C.2 Tests on the transition parameters

Table C.4: Significance of the market HMM transition parameters according to the reference regime

Transition	Ref. Regime 0			Ref. Regime 1			Ref. Regime 2			Ref. Regime 3		
	Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.
R0 → R0	-	-	-	2.8145	0.0001	***	3.2104	0.0000	***	2.2864	0.0002	***
R1 → R1	3.8162	0.0000	***	-	-	-	4.3308	0.0000	***	4.2535	0.0000	***
R2 → R2	3.3155	0.0018	**	2.4354	0.0004	***	-	-	-	3.4029	0.0045	***
R3 → R3	2.8188	0.0000	***	3.8467	0.0003	***	4.1372	0.0000	***	-	-	-
R0 → R1	-2.8145	0.0001	***	-3.8162	0.0000	***	-3.3155	0.0018	**	-2.2864	0.0002	***
R0 → R2	-3.2104	0.0000	***	-0.3958	0.7287		-3.3155	0.0018	**	-0.9240	0.3297	
R0 → R3	-2.2864	0.0002	***	0.5282	0.5983		-3.4029	0.0046	**	-0.5282	0.5983	
R1 → R2	-0.5146	0.6544		-4.3308	0.0000	***	-2.4354	0.0004	***	-0.0773	0.9502	
R1 → R3	-0.4373	0.6936		-4.2535	0.0000	***	-3.4029	0.0046	**	-0.0773	0.9502	
R2 → R3	-0.0875	0.9599		-0.9676	0.5171		-3.4029	0.0046	**	-4.1372	0.0000	***
R3 → R0	-1.0278	0.3832		1.0278	0.3831		1.3184	0.2268		-2.8188	0.0000	***

Reading example: The coefficient $R1 \rightarrow R1$ (reference Regime 0) is 3.8162 ($p = 0.0000$). This positive and highly significant (***) value indicates that, once established in Regime 1 (Expansion), the market has a far higher log-probability of remaining in it than of switching to Regime 0 (Flight-to-Quality). This result is consistent with the high average duration of Regime 1 (21.25 months) observed in the persistence diagnostics.

Table C.5: Significance of the pure macroeconomic HMM transition parameters according to the reference regime

Ref.	Transition	Coef.	Std. Err.	p-value	Sig.
R0	$R0 \rightarrow R1$	-4.002	0.715	< 0.001	***
R0	$R0 \rightarrow R2$	-4.690	1.005	< 0.001	***
R0	$R1 \rightarrow R1$	4.813	0.580	< 0.001	***
R0	$R1 \rightarrow R2$	-1.099	1.155	0.341	
R0	$R2 \rightarrow R1$	32.994	24.381	0.176	
R0	$R2 \rightarrow R2$	36.812	24.381	0.131	
R1	$R0 \rightarrow R0$	4.002	0.715	< 0.001	***
R1	$R0 \rightarrow R2$	-0.688	1.228	0.575	
R1	$R1 \rightarrow R0$	-4.813	0.580	< 0.001	***
R1	$R1 \rightarrow R2$	-5.912	1.002	< 0.001	***
R1	$R2 \rightarrow R0$	-32.994	0.000	< 0.001	***
R1	$R2 \rightarrow R2$	3.819	0.584	< 0.001	***
R2	$R0 \rightarrow R0$	4.690	1.005	< 0.001	***
R2	$R0 \rightarrow R1$	0.688	1.228	0.575	
R2	$R1 \rightarrow R0$	1.099	1.155	0.341	
R2	$R1 \rightarrow R1$	5.912	1.001	< 0.001	***
R2	$R2 \rightarrow R0$	-36.812	0.000	< 0.001	***
R2	$R2 \rightarrow R1$	-3.819	0.584	< 0.001	***

Reading example: For a transition from Regime 1 (Normal Regime), the coefficient towards Regime 0 (Crisis) is -4.813 ($p < 0.001$). This negative and significant value indicates that the probability of switching towards a crisis is very low compared with remaining in the normal state. Conversely, the coefficient $R2 \rightarrow R2$ (reference R1) is 3.819 ($p < 0.001$), confirming the strong persistence of the inflationary stress regime once it is established.

Table C.6: Significance of the HMM-TVTP transition parameters according to the reference regime (Intercepts)

Transition	Ref. Regime 0			Ref. Regime 1			Ref. Regime 2			Ref. Regime 3		
	Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.
$R0 \rightarrow R0$	-	-	-	3.2334	0.0000	***	5.7111	0.0060	**	3.0011	0.0000	***
$R1 \rightarrow R1$	4.8849	0.0000	***	-	-	-	6.6366	0.0005	***	5.7422	0.0000	***
$R2 \rightarrow R2$	3.6440	0.0124	*	2.5657	0.0051	**	-	-	-	8.0529	0.0748	
$R3 \rightarrow R3$	2.3692	0.0000	***	7.2506	0.1413		4.1836	0.0006	***	-	-	-
$R0 \rightarrow R1$	-3.2334	0.0000	***	-4.8849	0.0000	***	2.4778	0.2557		-0.2322	0.8146	
$R0 \rightarrow R2$	-5.7111	0.0060	**	-2.4778	0.2559		-3.6440	0.0125	*	-2.7100	0.2204	
$R0 \rightarrow R3$	-3.0011	0.0000	***	0.2322	0.8146		2.7100	0.2202		-2.3692	0.0000	***
$R1 \rightarrow R2$	-1.7517	0.3858		-6.6366	0.0005	***	-2.5657	0.0051	**	-0.8944	0.6809	
$R1 \rightarrow R3$	-0.8573	0.5671		-5.7422	0.0000	***	0.8944	0.6814		-7.2506	0.1497	
$R2 \rightarrow R3$	-4.4089	0.3579		-5.4873	0.2370		-8.0529	0.0818		-4.1836	0.0006	***
$R3 \rightarrow R0$	-2.3692	0.0000	***	4.8814	0.3234		1.8145	0.1638		-2.3692	0.0000	***

Reading example: For an intrinsic transition (without macroeconomic influence), the coefficient $R1 \rightarrow R1$ with Regime 0 as the reference is 4.8849 ($p = 0.0000$). This positive and highly significant (***) coefficient indicates that the probability of remaining in Regime 1 is far higher than that of transitioning towards Regime 0. In general, the high values on the diagonal ($Ri \rightarrow Ri$) confirm the strong temporal persistence of each identified regime.

Table C.7: Significance of the HMM-TVTP macroeconomic weights (W) by reference regime

Variable	Target	Ref. Regime 0			Ref. Regime 1			Ref. Regime 2			Ref. Regime 3		
		Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.	Coef	p-val	Sig.
Inflation	R0	-	-	-	-0.282	0.357		-0.818	0.068	.	0.918	0.067	.
	R1	0.282	0.357		-	-	-	-0.536	0.189		1.200	0.028	*
	R2	0.818	0.068	.	0.536	0.189		-	-	-	1.737	0.006	**
	R3	-0.918	0.067	.	-1.200	0.028	*	-1.737	0.006	**	-	-	-
Spread	R0	-	-	-	-0.186	0.547		-0.651	0.103		0.668	0.276	
	R1	0.186	0.547		-	-	-	-0.465	0.184		0.854	0.168	
	R2	0.651	0.103		0.465	0.184		-	-	-	1.318	0.044	*
	R3	-0.668	0.276		-0.854	0.168		-1.318	0.044	*	-	-	-
Slope	R0	-	-	-	0.979	0.003	***	0.068	0.914		-0.271	0.489	
	R1	-0.979	0.003	***	-	-	-	-0.912	0.165		-1.250	0.002	**
	R2	-0.068	0.914		0.912	0.165		-	-	-	-0.339	0.623	
	R3	0.271	0.489		1.250	0.002	**	0.339	0.623		-	-	-

Reading example: For a transition from **Regime 3** (reference), the coefficient of **Inflation** towards the target **Regime 2** is 1.737 ($p = 0.006$). This means that an acceleration of inflation significantly increases the probability of leaving a Flight-to-Quality state ($R3$) to enter an Inflationary Stress state ($R2$). Conversely, the coefficient towards Regime 1 is 1.200 ($p = 0.028$), indicating that high inflation also favours the return towards normal growth relative to remaining in a deflationary crisis.

C.3 Average duration and stationary distribution of the regimes

Table C.8: Average duration and stationary distribution of the HMM regimes

Regime	Average duration (months)	Stationary probability	Probability of remaining
R0	5.9528	0.1637	0.8320
R1	21.2498	0.4615	0.9529
R2	7.3630	0.1113	0.8642
R3	11.3100	0.2635	0.9116

Table C.9: Average duration and stationary distribution of the pure macroeconomic HMM regimes

Regime	Average duration (months)	Stationary probability	Probability of remaining
R0	37.41	0.194	0.973
R1	93.36	0.645	0.989
R2	46.54	0.161	0.979

Table C.10: Average duration and stationary distribution of the HMM-TVTP regimes

Regime	Average duration (months)	Stationary probability	Probability of remaining
R0	8.2707	0.1919	0.8791
R1	31.0233	0.5760	0.9678
R2	7.4766	0.0976	0.8662
R3	6.4715	0.1345	0.8455

C.4 Statistical scores

Table C.11: Robustness report and validation scores of the HMM model

Validation dimension	Score obtained	Critical threshold	Status
Statistical separability (s_{SE})	72.2 %	50.0 %	Valid
Numerical stability (s_{ST})	100.0 %	Positive Definite	Valid
Temporal persistence ($\mathbb{E}[D]$)	6.0 months	3.0 months	Valid
Decoding conviction (s_{CI})	72.7 %	70.0 %	Solid
Global Coherence Score (s_{RB})	80.56 %	60.0 %	Robust

Table C.12: Robustness report and validation scores of the pure macroeconomic HMM model

Validation dimension	Score obtained	Critical threshold	Status
Statistical separability (s_{SE})	50.0 %	45.0 %	Valid
Numerical stability (s_{ST})	100.0 %	Positive Definite	Valid
Temporal persistence ($\mathbb{E}[D]$)	37.4 months	3.0 months	Valid
Decoding conviction (s_{CI})	98.4 %	70.0 %	Solid
Global Coherence Score (s_G)	78.1 %	60.0 %	Robust

Table C.13: Robustness report and validation scores of the HMM-TVTP model

Validation dimension	Score obtained	Critical threshold	Status
Statistical separability (s_{SE})	72.2 %	50.0 %	Valid
Numerical stability (s_{ST})	100.0 %	Positive Definite	Valid
Temporal persistence ($\mathbb{E}[D]$)	6.5 months	3.0 months	Valid
Decoding conviction (s_{CI})	87.2 %	70.0 %	Solid
Global Coherence Score (s_{RB})	82.84 %	60.0 %	Robust

D Appendices to the dynamic allocation section

D.1 Definition of Risk-Budgeting

Given a covariance matrix Σ and risk budgets b_i for each asset i :

Problem formulation We consider the following objective function:

$$\min_{w>0} \quad \frac{1}{2} w^\top \Sigma w - \sum_{i=1}^n b_i \log(w_i)$$

where:

- the term $\frac{1}{2} w^\top \Sigma w$ corresponds to the variance of the portfolio,
- the term $-\sum_i b_i \log(w_i)$ acts as a logarithmic penalty ensuring the positivity of the weights and imposing a *risk budgeting*-type structure.

Constraints The problem is solved under a positivity constraint:

$$w_i > 0 \quad \forall i$$

The normalisation constraint $\sum_i w_i = 1$ is not imposed directly in the optimisation problem in order to facilitate numerical convergence, but is applied *a posteriori*:

$$w_i \leftarrow \frac{w_i}{\sum_j w_j}$$

Interpretation The first-order condition of the problem gives:

$$(\Sigma w)_i = \frac{b_i}{w_i}$$

which implies that the marginal contribution to risk of each asset satisfies:

$$w_i(\Sigma w)_i \propto b_i$$

Thus, each asset contributes to the total risk of the portfolio in proportion to its target budget.

Numerical implementation Since the problem is strictly convex on $\mathbb{R}_+^n$, it is solved efficiently using the L-BFGS-B algorithm, which makes it possible to impose the positivity constraints on the weights directly.

D.2 Definition of the financial metrics used

Given the monthly log-returns of the portfolio r_t , the following metrics are computed up to the final date T :

- **Annualised return:** the annualised geometric return is computed from the final wealth of the portfolio:

$$Return = Wealth_T^{\frac{12}{N}} - 1$$

where $Wealth_T$ represents the final cumulative value of the portfolio and N the total number of monthly observations.

- **Annualised volatility:** the volatility is computed from the simple monthly returns reconstructed from the log returns:

$$R_t = e^{r_t} - 1$$

then:

$$Vol = \sigma(R_t)\sqrt{12}$$

where $\sigma(R_t)$ denotes the standard deviation of the monthly returns.

- **Sharpe ratio:** the risk-adjusted performance is defined by:

$$Sharpe = \frac{12 \times (\mu^R - \mu^F)}{Vol}$$

where μ^R and μ^F represent respectively the means of the monthly returns of the portfolio and of the risk-free rate (r_t^F , approximated by the TB3MS from FRED (Federal Reserve Bank of St. Louis, 2026)).

- **Drawdown:** the relative loss with respect to the historical high is measured by:

$$DD_t = \frac{Wealth_t}{\max_{s \leq t}(Wealth_s)} - 1$$

- **Maximum Drawdown (MDD):** the maximum loss observed over the period is:

$$MDD = \min_t(DD_t)$$

- **Calmar ratio:** it relates the annualised return to the extreme-loss risk:

$$Calmar = \frac{Return}{|MDD|}$$

D.3 Macroeconomic explainability: contribution analysis via *Integrated Gradients*

In the framework of an inhomogeneous Markov chain (HMM-TVTP), the transition matrix is no longer constant but dependent on the macroeconomic covariates X_t . Consequently, any significant change in the transition probabilities must be explained by the evolution of the macroeconomic context.

In other words, we seek to answer the following question: "How can we explain the gap between the probability of switching towards a stress regime computed on the basis of yesterday's information and the one updated with today's new macroeconomic data?". For this, we use the *Integrated Gradients* (IG) method.

Mathematical formulation The objective is to explain the variation $\Delta P_{i,j}$ of the transition probability from state i towards state j between two successive dates:

$$\Delta P_{i,j} = P(z_{t+1} = j \mid z_t = i, X_{t+1}) - P(z_{t+1} = j \mid z_t = i, X_t) = p_{ij}(X_{t+1}) - p_{ij}(X_t)$$

The contribution of covariate m is computed as the integral of the gradient along an interpolation path going from X_t to X_{t+1} :

$$IC_m = (X_{t+1,m} - X_{t,m}) \cdot \int_{\alpha=0}^1 \frac{\partial p_{ij}(X_t + \alpha(X_{t+1} - X_t))}{\partial x_m} d\alpha$$

In practice, this integral is approximated numerically by a discrete sum (rectangle method):

$$IC_m \approx (X_{t+1,m} - X_{t,m}) \cdot \frac{1}{N} \sum_{k=1}^N \frac{\partial p_{ij}(X_t + \frac{k}{N}(X_{t+1} - X_t))}{\partial x_m}$$

D.4 Prediction of the volatilities and correlation

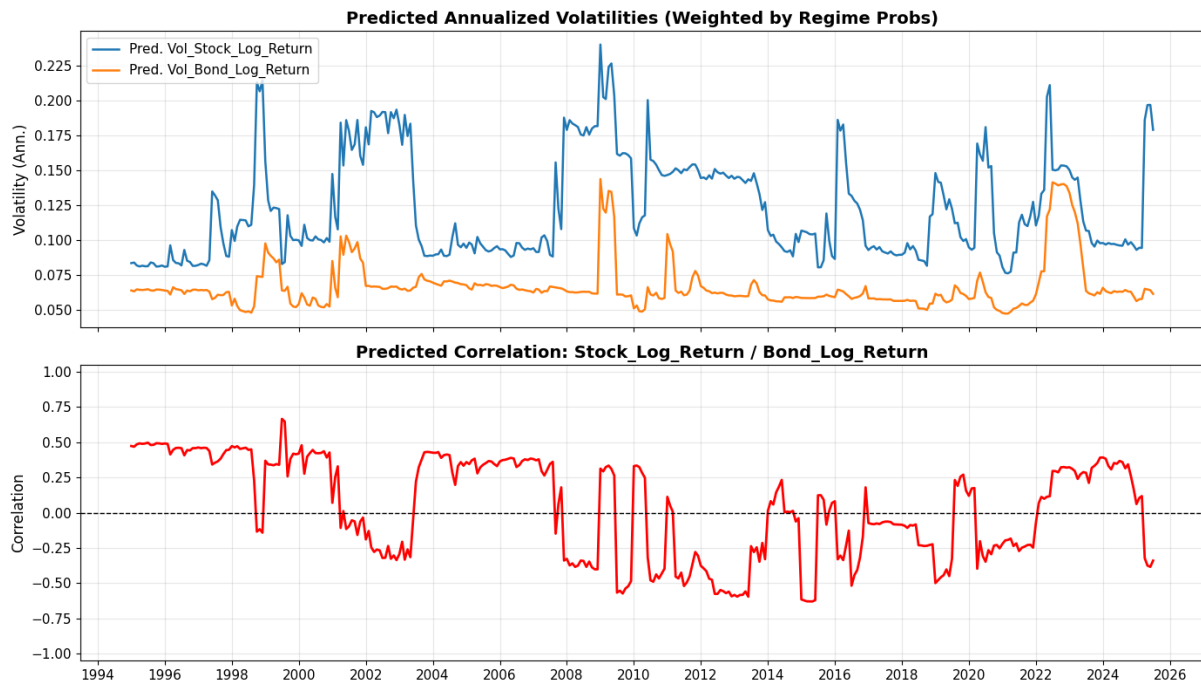


Figure D.1: Predicted volatilities and correlation during the *expanding* backtest 1995-2025

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